

scReg-Eval: a fixed-panel audit of regulatory alignment in single-cell RNA foundation-model gene graphs

Zeyu Fu

State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury,
Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine,
Army Medical University, Chongqing, China
Correspondence: fuzeyu99@126.com

1 *Frontiers in Genetics* | Original Research | Computational Genomics

2 Main figures: 13 | Appendix figures: 3 | Tables: 8

3 Abstract

4 **Background.** Claims that single-cell RNA foundation models encode gene regulation are
5 difficult to audit because many reference networks share co-expression structure with the
6 models' training data.

7 **Methods.** We introduce **scReg-Eval**, a fixed-panel audit protocol and software cap-
8 sule that compares frozen pairwise foundation-model gene graphs with a sequence- and
9 accessibility-derived regulatory-potential proxy while controlling co-expression, target peak
10 count, gene length, GC content, RNA detection rate, transcription-factor out-degree, and
11 target in-degree. The protocol-frozen panel contains 1,200 genes, including 446 in-range tran-
12 scription factors (TFs). We evaluate thirteen pooled model/readout combinations across
13 brain and paired peripheral blood mononuclear cell (PBMC) multiome data using two
14 plus-one Monte Carlo randomization tests ($N = 999$ each): a gene-label permutation
15 and a within-TF-row, degree-preserving target shuffle. Benjamini–Hochberg correction is
16 applied within eight prespecified tissue-by-confound-specification-by-null families. The es-
17 timand is fixed-panel unusualness of alignment under named confounds and two named
18 randomizations—not causal TF–target recovery and not model ranking.

19 **Results.** Dual-null “Support” under the primary full confound specification is seven of
20 thirteen rows (six small positive alignments, ρ from 0.00425 to 0.01241, plus one negative
21 Geneformer attention row in brain), but that count is not regulatory recovery: the same
22 seven rows appear only when degree covariates computed from the proxy are conditioned
23 on; under the co-primary non-degree specification dual-null Support is 0/13 and three of
24 the six positives change sign; and five of the six positive dual-null rows fail the protocol's
25 own FM–co-expression concordance check (Spearman -0.18 to -0.02), while PBMC scGPT
26 weakly passes (Spearman $\approx +0.01$) yet still fails protocol-pass (e.g. non-degree same-sign;
27 protocol-pass 0/13 overall), so dual-null Support marks unusual residual geometry under
28 disclosed degree hazard, not recovered regulation. The co-expression baseline, tested on the
29 same edge set outside FM BH families (uncorrected p), is itself dual-null significant in both
30 tissues, and the PBMC Geneformer-embedding alignment ($\rho = 0.00425$) is smaller than its
31 own baseline ($\rho = 0.0070$). A supervised TF-disjoint probe on paired PBMC data gives
32 mixed family-level results: UCE versus co-expression is supported by paired sign-flip testing
33 ($q = 0.0325$), as is the random-init floor ($q = 0.0300$), whereas Geneformer embedding
34 ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not. The probe
35 therefore must not be summarized as showing that all models are indistinguishable from
36 co-expression.

37 **Conclusions.** The audit finds weak, readout-specific, specification-dependent residual
38 alignment concentrated in embedding-geometry readouts. It neither supports a global
39 regulatory-capability claim for these scFM gene graphs nor licenses a field-level negative
40 conclusion beyond this fixed-panel protocol instance.

Keywords: single-cell RNA-seq, foundation models, gene regulatory networks, co-expression confounding, randomization inference, benchmark audit

1 Introduction

Single-cell RNA foundation models (scFMs) are increasingly used as frozen representations for network inference and perturbation analysis (1, 2). A high score against a curated or transcriptome-derived regulatory network, however, need not imply that a model encodes regulation beyond co-expression (3–5). Both the reference and the model can inherit gene abundance, detectability, hubness, and co-variation, so agreement with a co-expression-laden reference can look like regulatory recovery when it is mostly shared structure. Standardized scFM task suites and living GRN benchmarks have begun to systematize evaluation (5–7), but most still score expression-derived or literature GRNs rather than separating edge weights from expression statistics in the reference construction. Without that separation, it is hard to tell whether an scFM gene graph carries residual regulatory alignment after co-expression and simple graph-degree structure are controlled.

We address this problem as an audit-design question rather than as a model-ranking exercise. scReg-Eval freezes a transcription-factor (TF)–target panel, constructs an accessibility-and-motif regulatory-potential proxy whose *edge weights* are assembled without expression statistics, applies the same edge mask and confounds to every model, and evaluates the observed alignment against two structurally different randomization nulls. Panel membership and ATAC peak presence remain expression-adjacent design choices, so the proxy is not claimed to be RNA-structure-independent in every upstream decision—only that its scored edges are not co-expression weights. This design does not make the proxy causal ground truth. It asks a narrower, reproducible question: on the fixed panel, does a model gene graph align with the proxy after controlling measured RNA and graph structure, and is that alignment unusual under both label and target-assignment randomizations? Negative benchmarks of foundation-model interpretability under careful controls have precedent (8–10); our aim is the audit protocol, software capsule, and reporting discipline rather than a global verdict that scFMs do or do not encode regulation. Failure to support a global capability claim on this panel equally cannot be read as a field-level negative beyond this protocol instance.

The closest prior results show that foundation-model attention recovers co-expression rather than unique regulatory signal (11), and that residual-stream features of Geneformer and scGPT likewise encode organized pathway and interaction knowledge with minimal causal regulatory logic (12). Our reference differs from transcriptome-derived networks used in those interpretability studies: the proxy edge weights here are built from genomic sequence (motif matches) and chromatin accessibility only, so shared expression statistics alone cannot satisfy the reference through co-expression edges. That construction choice is what makes the audit informative when alignments are small: residual agreement cannot be explained by shared expression edge weights alone, but it also cannot be read as causal TF–target truth. Within one model, we additionally find that two readouts of the same frozen network disagree in sign (Geneformer embedding +0.0044 versus attention –0.0036 in brain)—dual-null Support on both sides of zero that is not regulatory recovery and not a model-level capability claim—a per-readout inconsistency that single-readout studies cannot see.

Our contributions are:

1. a reusable fixed-panel audit protocol and hash-pinned software capsule (panel manifests, null semantics, FDR families, and one-command validation) for scFM gene-graph claims, with biology serving as the application scene rather than as a claim of recovered regulatory truth;
2. a sequence/accessibility regulatory-potential proxy and 446-TF by 1,200-gene panel that

show moderate observed agreement across brain, blood, and a mixed fibroblast reprogramming dataset (motif/proximity stability is not a causal ceiling);

3. a full-confound partial-Spearman statistic that controls co-expression and six structural covariates, with a non-degree specification reported as a co-primary sensitivity analysis;
4. two finite-panel Monte Carlo tests with explicit null semantics, deterministic seed streams, plus-one correction, and eight prespecified FDR families;
5. a multi-readout application of Geneformer, scGPT, scFoundation, and UCE (1, 2, 13–15)—purely transcriptomic FMs, as distinct from knowledge-informed models that inject prior GRN structure at pretraining (16, 17)—showing small, readout-specific primary alignments—including a sign-inconsistent embedding/attention pair within one model—rather than a uniform positive or negative result; and
6. descriptive per-cell-type, cross-tissue, and axis-aligned injection diagnostics whose scope is kept separate from the pooled inferential families.

These choices are reported as ordinary design rules rather than as a separate boxed checklist. We freeze and hash the TF–target panel; label the reference as a regulatory-potential proxy rather than causal truth; control co-expression and structural covariates; state exactly what each randomization changes; use plus-one Monte Carlo p -values; define FDR families before reading results; and keep exploratory cell-type and sensitivity analyses distinct from the primary test. Relative to scFM interpretability studies, living GRN benchmarks, and scFM task suites (Table 1), scReg-Eval is designed to *complement* rather than compete: it holds motif/ATAC edge weights free of expression statistics, freezes dual-null families, and ships a hash-pinned audit capsule rather than a leaderboard score. Methods define the panel, proxy, confounds, and nulls; Results report the primary full-confound audit and the co-primary non-degree sensitivity; Discussion and the final figure set bound what the alignments can and cannot claim. Figure 1 states the two-tissue pairing, the ± 2 kb linkage rule, and the frozen panel before the audit heatmaps.

2 Methods

2.1 Estimand

The target of inference is *fixed-panel unusualness*: on one protocol-frozen TF–target panel, is the partial-Spearman alignment between a frozen foundation-model gene graph and the accessibility/motif proxy unusually large (in absolute value) under (i) a named full or non-degree confound design and (ii) two named randomizations (gene-label permutation and within-TF-row degree-preserving target shuffle), after within-family BH control? Dual-null “Support” answers that panel-local question only. Table 3 Support is full-spec / degree-hazard-conditioned by definition. It is not an estimand for causal TF–target effects, not a ranking of foundation models, and not a test of residual regulatory content after every possible confound. Concordance checks against co-expression further bound whether a dual-null row resembles attention-like co-expression geometry; they do not adjudicate residual regulatory truth. Discussion returns to this scope when interpreting Support counts. Reporting gates (dual-null Support, protocol-pass, FM-versus-baseline observed $\Delta\rho$, and probe contrast families) are frozen in the in-repo statistical analysis plan (SAP-v1.0; shipped with the release capsule). Digests for that plan live in the online deposit, not in the main text.

Table 1: Protocol axes versus related evaluation resources. Entries are qualitative design features, not performance ranks. scReg-Eval is a fixed-panel audit capsule; GeneRNIB and BioLLM-style suites score broader tasks against often expression-linked references.

Resource	Expression in reference?	Degree / confound control?	Explicit null semantics?	Multi-readout?	Hash-pinned capsule?
scReg-Eval (this work)	No (edge weights motif+ATAC)	Yes (coexp + 6 structural)	Yes (two MC nulls + BH families)	Yes	Yes
Kendiukhov et al. scFM interpretability	Yes (co-expression / pathways)	Partial	Varies	Attention-focused	No
Kendiukhov et al. SAE residual stream	Pathway/interaction features	Partial	Varies	Feature-level	No
GeneRNIB living GRN benchmark	Often literature / expression GRNs	Task-dependent	Benchmark scores	Multi-method	Community leaderboard
BioLLM / scFM task suites	Task labels from biology	Task-dependent	Standard ML splits	Multi-task	Code releases

2.2 Regulatory-potential proxy and fixed panel

The panel contains 1,200 genes, including 446 unique in-range transcription factors, protocol-frozen in-repo by a hash-pinned panel manifest in the audit capsule (exact digests are recorded in the deposit MANIFEST, not repeated here). This is an in-repository protocol freeze, not an external preregistration (OSF/AsPredicted) link. Panel construction further requires each gene to have at least one brain snATAC peak in the same ± 2 kb promoter-plus-gene-body window used for linkage, together with RNA-detection and vocabulary filters; the panel is therefore *not* ATAC-orthogonal. Accessible peaks from brain snATAC-seq, paired PBMC multiome, and a chemical reprogramming ATAC dataset (GSE206767, in which day-0 BJ fibroblasts are pooled with day-3 induced neuron-like cells as a single ALL accessibility track) are linked to genes by that window rule, which retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%) and systematically excludes distal enhancer elements. JASPAR2024 CORE vertebrate motifs are scanned with MOODS (18, 19); motif evidence in an accessible peak contributes a directed TF-target regulatory-potential weight, following the motif-in-accessible-region linkage paradigm used by regulon-inference methods (20). At the $p < 10^{-5}$ motif threshold (500 bp accessible windows, both strands, 554 motifs), the expected random hit count is 5.5 per peak. Edge weights are assembled without expression statistics; panel membership and ATAC peak-presence filters remain expression-adjacent inputs, so the proxy is not claimed RNA-structure-independent in every design choice. Accessibility, motif occurrence, and proximity still do not establish direct regulation or causality. We therefore call it a proxy throughout. The two evaluation tissues differ in pairing: the PBMC data are a paired multiome assay (RNA and ATAC from the same cells), whereas the brain analysis combines snATAC-seq with cross-study RNA, so the co-expression control is intrinsically stronger in PBMC than in brain (Fig. 1A).

Brain analyses apply a prespecified neural marker mask to every brain readout; in this panel the mask removes 446 of 534,754 brain edges (0.08%), a small fixed-panel exclusion because only one of the 34 prespecified markers (VCAN, which is not a transcription factor) falls in the fixed

panel. PBMC edges are unmasked under the prespecified tissue policy. Before analysis, the driver verifies the manifest, TF identity and order across all caches, matrix dimensions and finiteness, and four pinned legacy-file hashes.

2.3 Foundation-model and baseline graphs

Gene graphs are read from cached, model-validated outputs over the same manifest. Available pooled readouts are Geneformer embedding, attention, and in-silico knockout; scFoundation encoder; UCE encoder (4-layer variant, checkpoint pinned by the pipeline); and scGPT encoder; plus a random-init Geneformer floor. Coverage differs by tissue because only precomputed, validated graphs are included: brain has eight FM/readout rows and PBMC has five. Co-expression is reported as a baseline on the same edge set and nulls, but is not included as an FM row or in FM multiplicity families. Co-expression enters its own test only as the signal, never as a covariate: its design matrix is the full specification minus the co-expression column—an intercept plus the six structural covariates (target peak count, gene length, RNA detection rate, GC content, TF out-degree, target in-degree)—because partialling co-expression out of itself is degenerate by construction. CellPLM is outside scope because its architecture does not expose contextual per-gene states (21). Inference settings differ by model and are reported rather than harmonized: PBMC readouts use a deterministic 4,000-cell pool for every model; brain Geneformer embedding and scGPT use all available cells, brain Geneformer attention is capped at 4,000 cells, brain UCE uses 2,000 cells, and brain scFoundation uses 1,500 cells with a 512-token context cap. Manifest gene coverage is 1,159/1,200 for PBMC UCE, 1,195/1,200 for brain UCE, 1,199/1,200 for brain scFoundation, 1,168/1,200 for PBMC scFoundation, and 1,200/1,200 otherwise.

2.4 Observed statistic and confound specifications

For each retained TF–target pair, all graph weights are rank transformed. The primary statistic is the Pearson correlation between residualized ranks, equivalent to partial Spearman correlation. The *full* design contains an intercept, ranked co-expression, and standardized target peak count, gene length, RNA detection rate, GC content, TF out-degree, and target in-degree. Target peak count is the count of *all* linked peaks in the gene window (no per-gene cap). GC content is estimated from at most 40 peaks per gene (centered windows on the peak midpoints), so GC understates the extremes of very peak-dense genes while peak count does not. The two degree covariates are computed from the proxy graph itself, so the full specification conditions on functions of the outcome; this is a validity hazard, not a bookkeeping detail, and it is why the *non-degree* design, which omits both degree columns, is reported co-primary rather than as a footnote. Observed and randomized statistics always use matching column sets. Writing $r(\cdot)$ for ranks and $x^{\perp C}$ for the residual of x after linear projection onto a named confound matrix C ,

$$\rho_{\text{spec}} = \text{corr}(r(G_{\text{FM}})^{\perp C_{\text{spec}}}, r(G_{\text{proxy}})^{\perp C_{\text{spec}}}), \quad (1)$$

where C_{full} is the full design and C_{nondeg} omits TF out-degree and target in-degree. Figures 4, 7, and 8 report this statistic.

2.5 Finite-panel randomization inference

The TF panel is treated as fixed; no TF-superpopulation bootstrap (22) or confidence interval is used. For the gene-label null, one permutation relabels both endpoints of the proxy graph, in the spirit of the Mantel matrix correspondence test (23). The FM graph, co-expression, and non-degree gene covariates remain fixed. In the full specification, proxy-derived TF out-degree and target in-degree are recomputed after relabeling.

The second null independently shuffles edge weights within each TF row across non-self targets. This preserves TF out-degree and the row’s weight multiset while changing target assignment; target in-degree is recomputed only for the full specification. One proxy randomization per replicate is shared across model rows, with each model scored against the same replicate. Explicit-index tests verify numerical equivalence to separate least-squares calculations, including rank-deficient designs.

Both tests use two-sided plus-one correction (24),

$$p_{\text{MC}} = \frac{1 + \sum_{b=1}^{999} \mathbf{1}\{|T_b| \geq |T_{\text{obs}}|\}}{1000}, \quad (2)$$

with resolution 0.001. Figure 6 uses this Monte Carlo p -value. Benjamini–Hochberg correction (25) is performed within each of eight families: tissue (brain/PBMC) $\times$ confound specification (full/non-degree) $\times$ null (gene-label/row-shuffle). The full specification is the protocol-frozen primary (in-repo family definitions); the non-degree specification is reported alongside it as co-primary because the full specification conditions on proxy-derived degree covariates. Dual-null Support on an FM row is the conjunction of both within-family BH thresholds,

$$\text{Support} \iff q_M < 0.05 \text{ and } q_D < 0.05, \quad (3)$$

with q_M the gene-label (Mantel) BH q and q_D the degree-preserving row-shuffle BH q . Figure 4 reports Support under this rule. A sign flip between specifications is

$$\text{sign}(\rho_{\text{full}}) \neq \text{sign}(\rho_{\text{nondeg}}). \quad (4)$$

Figure 7 marks these sign flips. Protocol-pass is the conjunction of dual-full Support, FM–co-expression concordance (Spearman > 0), non-degree same sign as the full estimate, multi-readout sign agreement within model family and tissue, and full-spec ρ strictly above the tissue co-expression baseline; dual non-degree Support is reported but not required for protocol-pass. Figure 12 applies these gates.

2.6 TF-disjoint supervised probe

The pooled audit is an unsupervised raw-readout test. As a supervised complement, run on paired PBMC multiome data only, we train a ridge probe to predict each TF’s proxy-target profile from the FM graph itself. The probe was originally specified over per-gene embeddings; because the validated caches store gene-by-gene graphs rather than embeddings, we reformulated it at the pair level, which answers the same question without new model inference. Samples are ordered (TF, gene) pairs; features for each family are the edge weight $S_{tf,g}$, its reverse $S_{g,tf}$, and the within-TF rank of the edge. These are scalar per-edge summaries, so the probe tests edge-level linear recoverability, not row-level regulatory programs. Nothing in the design is indexed by TF identity, so a probe trained on 311 held-in TFs transfers unchanged to 135 disjoint test TFs. Regularization is chosen by leave-one-out cross-validation over training pairs. Performance is the per-test-TF Spearman correlation between predicted and observed proxy profiles, averaged over test TFs, after residualizing both sides against target peak count, gene length, GC content, and detection rate. Writing C' for that four-column probe design and $\hat{y}_t$ for the ridge prediction on test TF t ,

$$\bar{\rho}_{\text{probe}} = \frac{1}{n_{\text{test}}} \sum_t \text{Spearman}(\hat{y}_t^{\perp C'}, y_t^{\perp C'}). \quad (5)$$

Figure 10 reports this mean and the paired sign-flip test of per-TF differences versus co-expression. The null permutes gene labels within each test TF; paired FM-versus-co-expression

contrasts use sign-flipped per-TF differences. Gene-level degree features are withheld from the primary probe arm because they let the probe reach the proxy through gene-level confounding rather than through the graph’s edges.

2.7 Descriptive and pipeline-sensitivity analyses

Per-cell-type rows report observed partial correlations, cell counts, ranges, medians, and sign counts. They have no randomization p -values or FDR adjustment. Cross-tissue proxy reproducibility is also observed-only. On the shared non-self TF–target edge set, the Mantel rank statistic between two graphs is

$$\rho_{\text{Mantel}}(G, H) = \text{Spearman}(\text{vec}(G), \text{vec}(H)). \quad (6)$$

Figures 2, 11, and A1 report this observed agreement. An additive TF-row-plus-gene-column fit on one tissue’s proxy,

$$\begin{aligned} \widehat{W}_{ij} &= \mu + a_i + b_j, \\ f_{\text{add}} &= \frac{\rho(\widehat{W}^{(A)}, W^{(B)})}{\rho(W^{(A)}, W^{(B)})}, \\ \rho_{\text{resid}} &= \text{Spearman}(W^{(A)} - \widehat{W}^{(A)}, W^{(B)} - \widehat{W}^{(B)}), \end{aligned} \quad (7)$$

is scored against the other tissue; f_{add} is the additive fraction and ρ_{resid} is rank agreement after removing each graph’s own additive fit. Figure 3 reports these quantities; Figs. A1–A3 reuse them on construct-lane overlays. These quantities are descriptive; they are not a new randomization test.

Concordance is

$$\rho_{\text{conc}} = \text{Spearman}(G_{\text{FM}}, G_{\text{coexp}}) \quad (8)$$

on the same edge set; a readout passes if $\rho_{\text{conc}} > 0$. Figure 5 plots this attention-likeness check; it does not adjudicate residual regulatory content. The Geneformer knockout (KO) readout is an in-silico perturbation of the same frozen graph; the artifact-corrected KO subtracts a random-token position-shift baseline and is not a positive control. An attention-omission guard repeats the brain dual-null decision with attention omitted and a separate seed stream; the comparison is Support decisions, not q -values.

Nested covariate-ladder rungs (marginal $\rightarrow$ co-expression $\rightarrow$ non-degree $\rightarrow$ degree $\rightarrow$ full) evaluate Eq. (1) at successive named C matrices. The sequence is descriptive and is not a model-selection criterion.

Finally, for each tissue we evaluate the following injected fractions:

$$\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}.$$

Writing r_{proxy} for proxy ranks residualized against C_{full} , the injected graph is

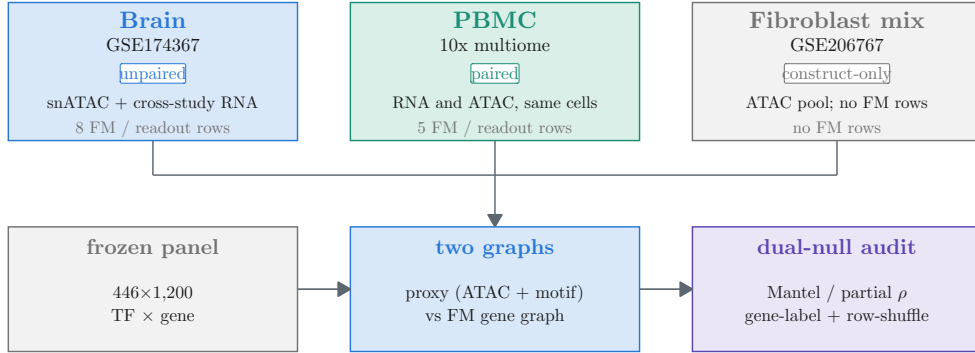
$$G_{\text{synth}}(\alpha) = \alpha z(r_{\text{proxy}}) + (1 - \alpha) z(\varepsilon), \quad (9)$$

where ε is independent standard Gaussian noise and the observed statistic is repeated 30 times per α . Figure 9 reports this ladder. The α -equivalent of an observed ρ is the ladder α whose mean recovered ρ matches that observation (descriptive interpolation, not an MDE). This axis-aligned diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

The fixed panel, model/readout availability, inference cell counts, and edge-accounting steps are audited explicitly in Fig. 13; blank coverage cells denote missing validated graphs, not negative results. The full audit therefore comprises 13 pooled FM/readout rows rather than a

A

Two-tissue design, one frozen panel



B

Linkage rule admits 4–6% of peaks

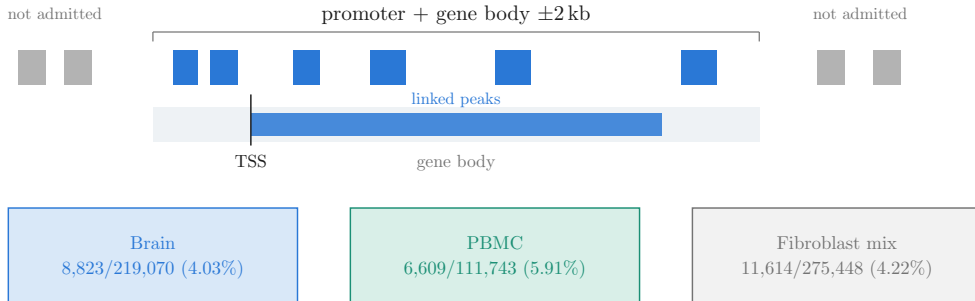


Figure 1: Two-tissue study design and the ± 2 kb linkage rule. (A) Unpaired brain snATAC (GSE174367) plus cross-study RNA versus paired PBMC 10x multiome RNA/ATAC; the fibroblast mix (GSE206767) is construct-only and contributes no foundation-model (FM) rows. All arms freeze the same $446 \times 1,200$ panel, then compare an accessibility/motif proxy graph with each FM gene graph under dual-null Mantel/partial ρ . (B) Peaks are admitted only inside a promoter-plus-gene-body ± 2 kb window: 8,823/219,070 (4.03%) in brain, 6,609/111,743 (5.91%) in PBMC, and 11,614/275,448 (4.22%) in the fibroblast mix. Distal and three-dimensional contacts are excluded by this rule.

272 balanced model-by-tissue factorial design.

273 3 Results

274 The study compares unpaired brain snATAC plus cross-study RNA with paired PBMC mul-
 275 tiome, holds a fibroblast ATAC mix as construct-only, and scores a frozen $446 \times 1,200$ panel
 276 under a promoter-plus-gene-body ± 2 kb linkage rule (Fig. 1).

277 3.1 The proxy is reproducible, but remains a proxy

278 The accessibility/motif construct shows moderate observed agreement across tissue consensus
 279 graphs: brain–PBMC $\rho = 0.5249$, brain–fibroblast-mix $\rho = 0.5245$, and PBMC–fibroblast-mix
 280 $\rho = 0.4623$ (Fig. 2; Table 2). The third dataset (GSE206767) is a chemical-reprogramming ex-
 281 periment whose accessibility track pools day-0 fibroblasts and day-3 induced neuron-like cells,
 282 so it does not represent any resting tissue. These are descriptive fixed-panel correlations without
 283 new randomization inference. They show the construct is stable across ATAC inputs of very
 284 different provenance, not a causal ceiling or proof of TF–target truth. Figure 2 also makes the

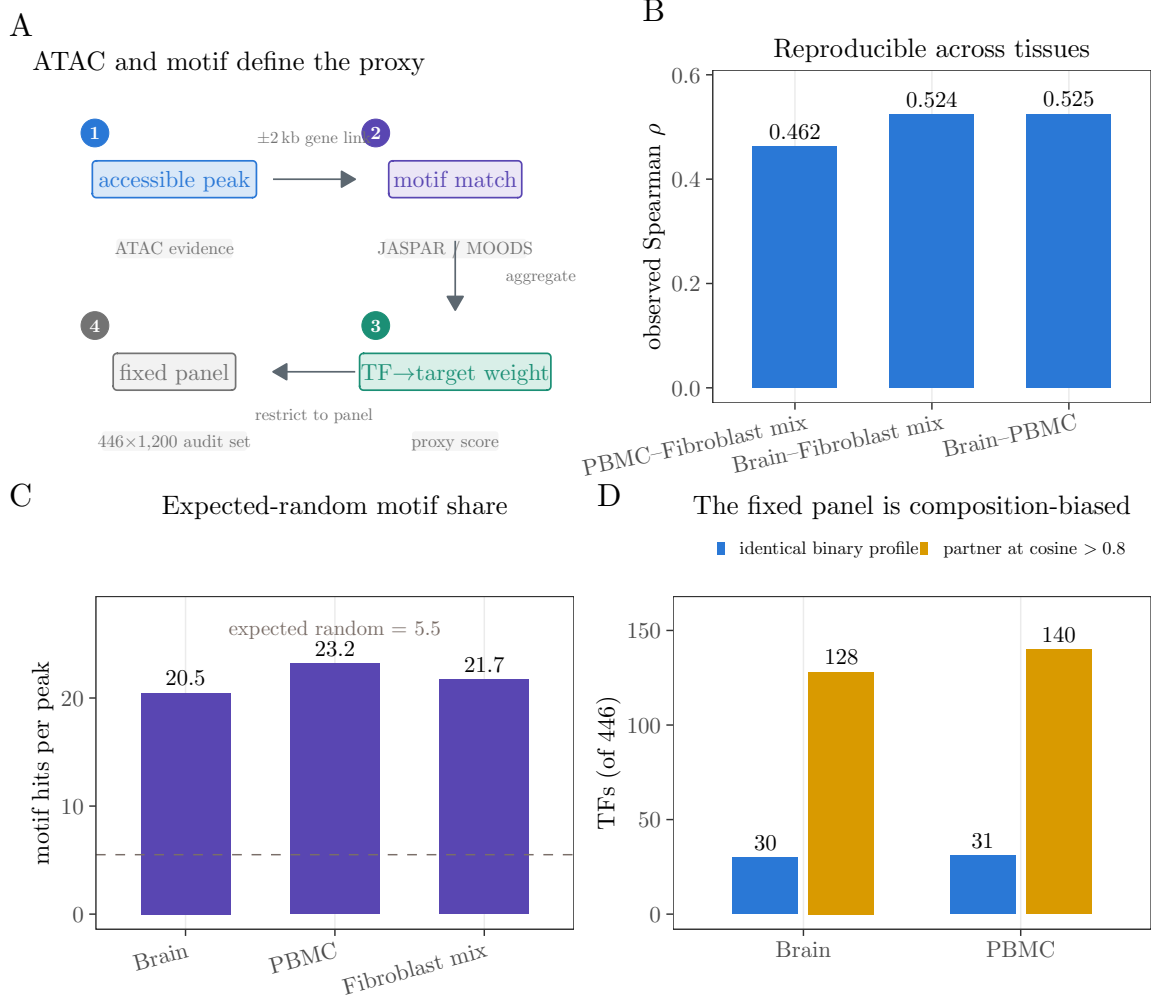


Figure 2: Regulatory-potential proxy, evidence, and panel composition. (A) Construction pipeline: accessible peaks are linked to genes (± 2 kb promoter-plus-gene-body), JASPAR/MOODS motif matches contribute directed TF→target weights, and edges are restricted to the hash-pinned 446-TF $\times$ 1,200-gene panel; edge weights are built without expression statistics (panel and ATAC inputs remain expression-adjacent), and the proxy is not causal ground truth. (B) Observed consensus-graph correlations across three tissue datasets. (C) Motif hits per accessible peak relative to the expected 5.5 random hits at the 10^{-5} motif threshold. (D) TF target-profile redundancy: counts with an identical binary profile or a partner at cosine similarity > 0.8 , out of 446 TFs.

proxy’s motif-noise and panel- composition limits explicit. Most of that stability is marginal structure: an additive TF-row-plus-gene-column model fit on one tissue’s proxy predicts the other tissue’s edges at $\rho = 0.32$ – 0.41 (69–78% of the observed agreement; Fig. 3), and after removing each tissue’s own additive structure the residual agreement is $\rho = 0.41$ – 0.55 . Table 2 reports the same quantities with edge Jaccard and mean per-TF-row Spearman. One pair deviates from that monotone reading: for brain–fibroblast-mix the additive fit is net-subtractive (residual $\rho = 0.55$ above the observed 0.52), so its additive-prediction fraction quantifies similarity in the marginals rather than decomposing the edge-level agreement. The dominant share of the apparent reproducibility is therefore fixed by the motif library and the proximity rule, both of which are tissue-independent by design, rather than by tissue-specific regulatory biology.

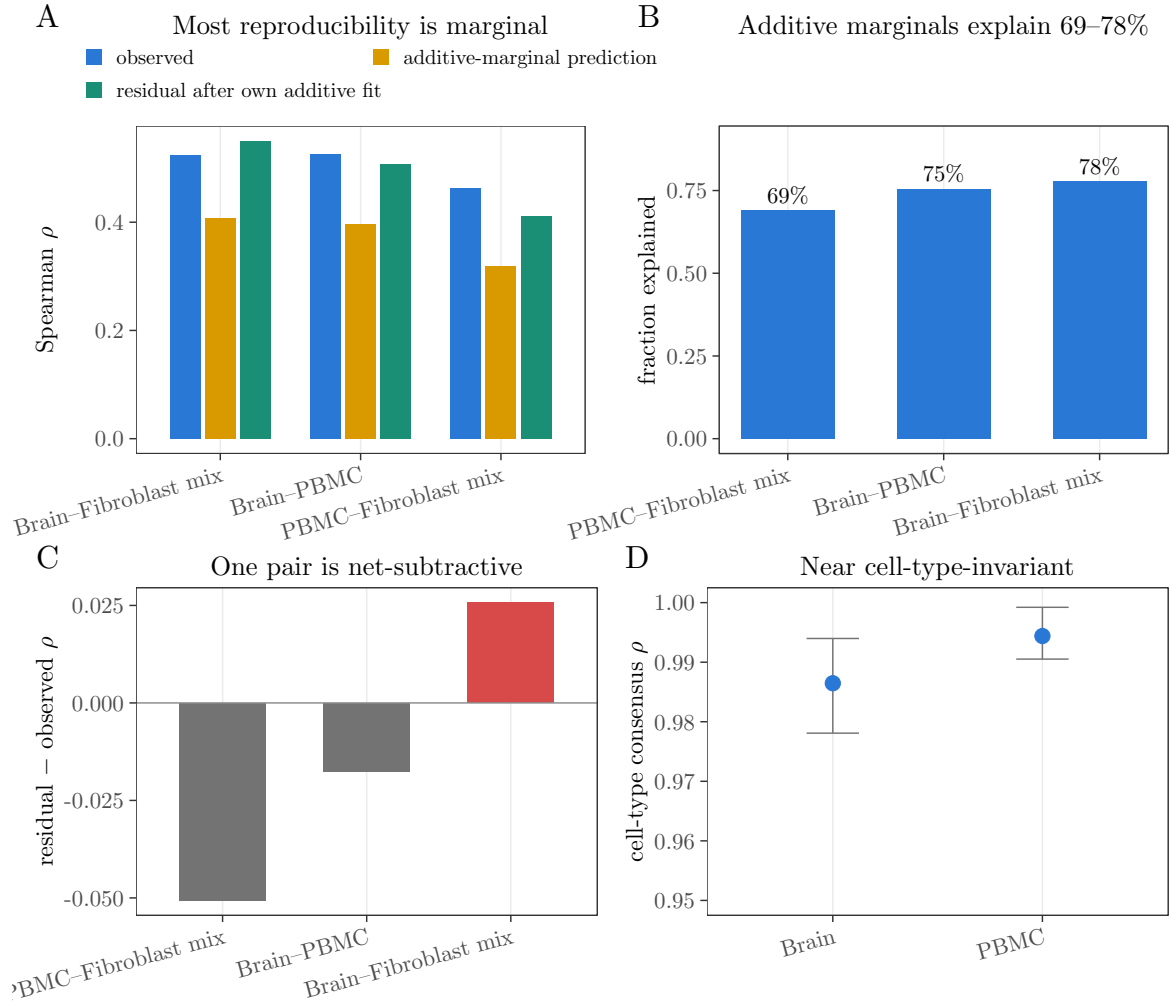


Figure 3: Cross-tissue structure of the regulatory-potential proxy on the frozen $446 \times 1,200$ panel (three tissue pairs). (A) Observed Spearman ρ , TF-row-plus-gene-column additive prediction, and residual ρ after removing each tissue’s own additive fit. (B) Fraction of observed agreement predicted by those additive marginals (69–78%). (C) Residual minus observed ρ ; red = net-subtractive (residual > observed), grey otherwise; brain–fibroblast mix is net-subtractive. (D) Mean pairwise cell-type proxy Spearman ρ within brain and PBMC; whiskers are the min–max range (not a confidence interval). These are descriptive fixed-panel quantities without new randomization inference.

Table 2: Observed cross-tissue reproducibility of the regulatory-potential proxy on the frozen $446 \times 1,200$ panel ($n = 3$ tissue pairs). Columns: observed Spearman ρ ; additive fraction of that agreement from TF-row plus gene-column marginals; residual Spearman after each tissue’s own additive fit; binary-support φ (phi coefficient of nonzero-edge presence/absence); edge Jaccard; mean per-TF-row Spearman; shared TF count (446). Fixed-panel descriptive quantities; no confidence interval or new randomization inference.

Tissue pair	Obs. ρ	Add. frac.	Resid. ρ	φ	Edge Jac.	Mean row ρ	Shared TFs
Brain–PBMC	0.5249	0.7543	0.5073	0.5070	0.416	0.4001	446
Brain–Fibroblast mix	0.5245	0.7771	0.5502	0.5004	0.414	0.4151	446
PBMC–Fibroblast mix	0.4623	0.6895	0.4117	0.4461	0.374	0.3364	446

3.2 Primary full-confound audit: six weak positives, one negative, and a baseline that matches them

Table 3 and Fig. 4 report all thirteen pooled FM/readout rows plus the co-expression baseline. Six small positive effects are supported by both randomization tests after within-family BH correction (Support column in Table 3); that full-spec Support count is authoritative only together with co-primary non-degree dual-null Support 0/13 and protocol-pass 0/13 (Table 4), not as a standalone recovery claim. In brain, Geneformer embedding has $\rho = 0.00442$ ($q_M = 0.006$, $q_D = 0.003$), UCE encoder has $\rho = 0.00461$ ($q_M = 0.004$, $q_D = 0.003$), and scGPT encoder has $\rho = 0.01241$ ($q_M = 0.004$, $q_D = 0.003$). In paired PBMC, Geneformer embedding has $\rho = 0.00425$ ($q_M = 0.002$, $q_D = 0.003$), scGPT encoder has $\rho = 0.01098$ ($q_M = 0.002$, $q_D = 0.003$), and UCE encoder has $\rho = 0.00593$ ($q_M = 0.002$, $q_D = 0.003$).

Table 3: Primary fixed-panel results under the *full* confound specification only (13 FM/readout rows plus the same-edge co-expression baseline). Partial Spearman ρ is alignment of each graph with the accessibility/motif proxy. M denotes the gene-label null and D the within-TF-row degree-preserving target shuffle ($N = 999$ each); each q is BH-adjusted within its tissue-by-specification-by-null family. Support is dual within-family BH $q_M < 0.05$ and $q_D < 0.05$ (“both”); opposite-sign dual support is labeled “both (neg)”; “neither” otherwise. The co-expression baseline is tested on the same edge set outside the FM families and is marked “–” in Support; its entries are uncorrected p -values (dashes mark where no family correction applies). Dual-null Support is not protocol-pass.

Tissue	Readout	Partial ρ	p_M	q_M	p_D	q_D	Support
Brain	Geneformer embed	0.00442	0.003	0.006	< 0.001	0.003	both
Brain	scFoundation encoder	0.00062	0.633	0.633	0.643	0.643	neither
Brain	UCE encoder	0.00461	< 0.001	0.004	< 0.001	0.003	both
Brain	scGPT encoder	0.01241	< 0.001	0.004	< 0.001	0.003	both
Brain	Geneformer attention	−0.00362	0.002	0.005	0.008	0.016	both (neg)
Brain	Random-init floor	0.00187	0.182	0.243	0.175	0.233	neither
Brain	Geneformer KO	−0.00093	0.407	0.465	0.366	0.418	neither
Brain	Artifact-corrected KO	−0.00189	0.100	0.160	0.091	0.146	neither
Brain	Co-expression baseline	0.0028	0.019	–	0.014	–	–
PBMC	Geneformer embed	0.00425	< 0.001	0.002	0.002	0.003	both
PBMC	Geneformer attention	−0.00018	0.866	0.866	0.839	0.839	neither
PBMC	scFoundation encoder	0.00112	0.393	0.491	0.359	0.449	neither
PBMC	scGPT encoder	0.01098	< 0.001	0.002	< 0.001	0.003	both
PBMC	UCE encoder	0.00593	< 0.001	0.002	< 0.001	0.003	both
PBMC	Co-expression baseline	0.0070	< 0.001	–	< 0.001	–	–

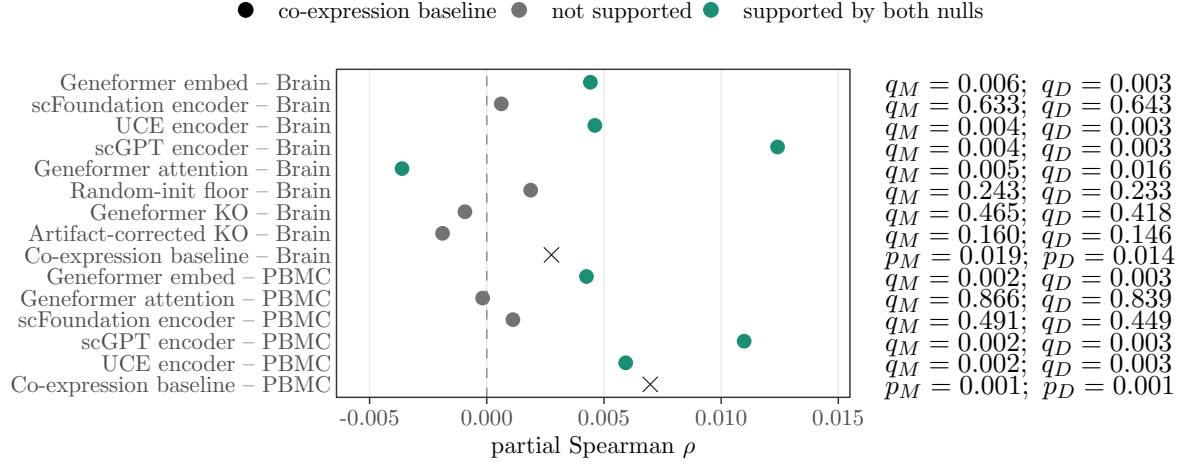
Dual-null Support is only one gate. Table 4 and Fig. 12 report six gate columns on the same thirteen FM rows (dual-null under full and under non-degree, FM–co-expression concordance (Spearman > 0), non-degree same sign as the full estimate, multi-readout sign agreement within model family and tissue, and full-spec ρ strictly above the tissue co-expression baseline). **Protocol-pass** is the conjunction of five of those gates (Dual non-deg is reported but not required): dual-full, concordance, non-degree same sign, multi-readout sign agreement, and $\rho > \text{baseline}$ simultaneously. Under those gates, 0/13 rows **protocol-pass**—including every dual-null positive—so dual-null Support is not treated as regulatory recovery (Fig. 12).

One further row is supported by both nulls with the opposite sign: brain Geneformer attention, $\rho = -0.00362$ ($q_M = 0.005$, $q_D = 0.016$), from the same model whose embedding readout is positive in the same tissue. A sign-inconsistent pair of readouts within one model is not what a regulatory-capability account predicts.

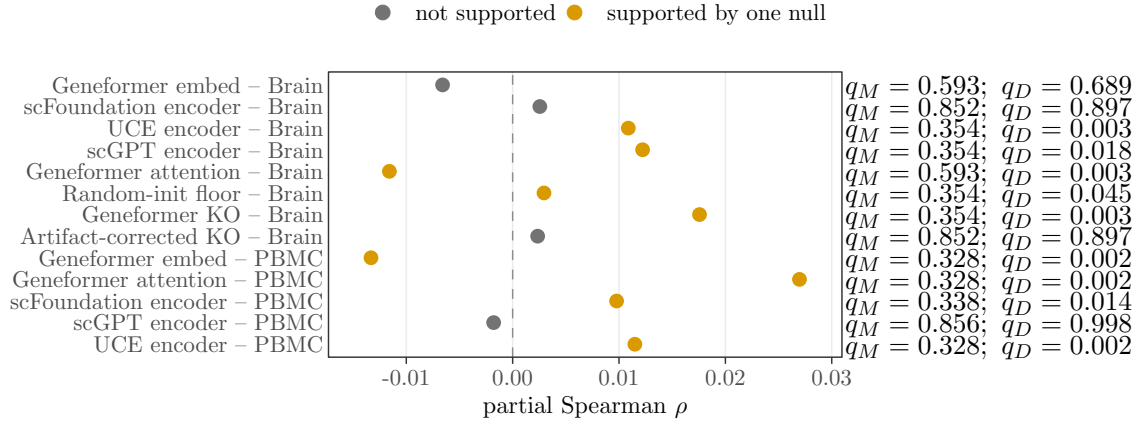
The remaining rows are not supported: brain scFoundation ($\rho = 0.00062$), Geneformer knockout (-0.00093) and its artifact-corrected knockout readout (-0.00189 ; the artifact-

A

Full-confound specification (primary)

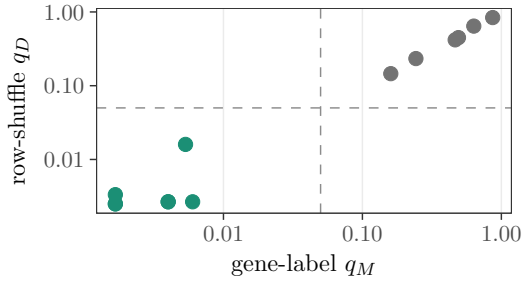


B

Non-degree co-primary (closest q shown; none below 0.05)

C

Both nulls below 0.05



D

Dual-null rows vs baseline

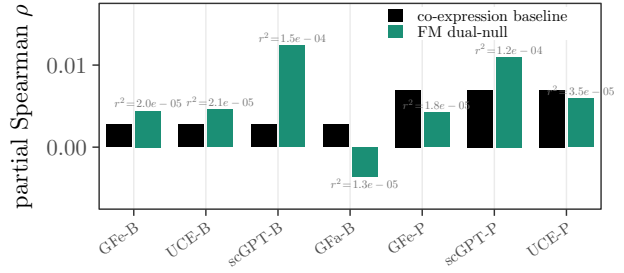


Figure 4: Primary fixed-panel audit of foundation-model (FM) gene graphs (Geneformer, Universal Cell Embedding (UCE), scGPT, scFoundation, and a random-init floor) under both prespecified confound specifications. (A) Full-confound pooled rows and the same-edge co-expression baseline; labels give Benjamini–Hochberg (BH) q_M (gene-label / Mantel) and q_D (degree-preserving row-shuffle) for FM rows and uncorrected p_M, p_D for the baseline. (B) Non-degree co-primary rows; no row is supported by both nulls. (C) Full-confound q_M versus q_D ; dual-null Support requires both coordinates below 0.05. (D) Dual-null rows versus the same-edge co-expression baseline; $r^2 = \rho^2$ is annotated on the FM bars (tiny effects; several dual-null positives do not exceed baseline).

Table 4: Protocol-pass matrix for the thirteen pooled FM/readout rows (yes = pass, no = fail). Dual full / Dual non-deg: both within-family BH $q_M, q_D < 0.05$ under the named confound specification. Concord.: FM-co-expression Spearman > 0 (attention-likeness). ND same sign: full and non-degree partial ρ share sign. Multi-RO sign: all full-spec readouts of the same model family in that tissue agree in sign (single-readout families pass). $\rho > \text{base}$: full-spec ρ exceeds the tissue co-expression baseline on the same edge set. Protocol-pass is the conjunction of Dual full, Concord., ND same sign, Multi-RO sign, and $\rho > \text{base}$ (Dual non-deg is reported but not required beyond same-sign, matching the dual-null failure mode under non-degree). All thirteen rows fail protocol-pass.

Tissue	Readout	Dual full	Dual non-deg	Concord.	ND same sign	Multi-RO sign	$\rho > \text{base}$	Protocol-pass
Brain	Geneformer embed	yes	no	no	no	no	yes	no
Brain	scFoundation encoder	no	no	yes	yes	yes	no	no
Brain	UCE encoder	yes	no	no	yes	yes	yes	no
Brain	scGPT encoder	yes	no	no	yes	yes	yes	no
Brain	Geneformer attention	yes	no	yes	yes	no	no	no
Brain	Random-init floor	no	no	no	yes	yes	no	no
Brain	Geneformer KO	no	no	no	no	no	no	no
Brain	Artifact-corrected KO	no	no	no	no	no	no	no
PBMC	Geneformer embed	yes	no	no	no	no	no	no
PBMC	Geneformer attention	no	no	yes	no	no	no	no
PBMC	scFoundation encoder	no	no	yes	yes	yes	no	no
PBMC	scGPT encoder	yes	no	yes	no	yes	yes	no
PBMC	UCE encoder	yes	no	no	yes	yes	no	no

corrected row subtracts a random-token position-shift baseline and is not a positive control—the genuine positive control in this audit is the signal-injection diagnostic below), the random-init Geneformer floor (0.00187), and PBMC Geneformer attention (−0.00018) and scFoundation (0.00112).

The co-expression baseline, run through the same nulls on the same edge set (it cannot enter the FM families because partialling co-expression out of itself is degenerate), meets both uncorrected null thresholds in both tissues (outside FM BH families; not FM-family “Support”): brain $\rho = 0.00276$ ($p_M = 0.019$, $p_D = 0.014$) and PBMC $\rho = 0.00697$ ($p_M = 0.001$, $p_D = 0.001$). Two observations follow. First, the baseline reaches the same order of alignment as the supported FM rows. Second, the PBMC Geneformer-embedding positive ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.00697$): on that row the model aligns with the proxy less than the confound it is adjusted against. Table 5 reports observed $\Delta\rho = \rho_{\text{FM,full}} - \rho_{\text{baseline}}$ for all 13 FM rows and a shared-null Mantel test of $\Delta > 0$ ($N=999$ gene-label draws, BH within tissue; `method=shared_mantel_delta_v1`): only four dual-null full rows beat their tissue baseline on the observed contrast (brain Geneformer embed, UCE, scGPT; PBMC scGPT), and only 2/13 rows are shared-null significant after BH (brain scGPT; PBMC scGPT), so dual-null Support does not imply multiplicity-controlled superiority to co-expression on the same edge set. Under an independence reference operating characteristic for dual-null Support (independent Uniform p_M, p_D , BH within each full-spec tissue family, 2×10^4 draws), the Monte Carlo rate of ≥ 1 dual-null row is $\approx 2 \times 10^{-4}$ (brain) and $\approx 3 \times 10^{-4}$ (PBMC), while the audit records 4 and 3 dual-null rows respectively—so dual-null Support is not explained by naive independent double-dipping alone, yet still is not regulatory recovery (release-capsule dual-null independence diagnostic).

Four completed enhancement-v1 checks qualify the fixed-panel interpretation without changing the frozen primary families. Five-fold cross-fitting of the proxy-derived degree covariates preserved the qualitative direction in all 13 pooled rows, with absolute changes from the frozen estimates no larger than 1.1×10^{-5} (E1). Linkage sensitivity preserved the edge-weight structure for TSS−1 kb and TSS−5 kb policies ($\rho = 0.988$ and 0.971 versus the baseline), whereas a promoter-only policy changed the construct substantially ($\rho = 0.119$; E2). In 20 fixed 80% TF subsamples, signs were concordant in all brain and PBMC draws; the descriptive ρ ranges were 0.00271–0.00610 and 0.00115–0.00716, respectively (E3). Detection-based stratification also showed larger alignment for the operational housekeeping proxy than for low-detection TF

Table 5: FM versus same-edge co-expression baseline under the full confound design. The ρ_{FM} column is a shared-null recompute of the same full-spec statistic; small differences from the primary fixed-panel ρ values reflect that recompute, not a second primary estimand. $\Delta\rho = \rho_{\text{FM}} - \rho_{\text{base}}$ is the observed contrast; q_{Δ} and Shared sig are one-sided shared-null Mantel tests of $\Delta > 0$ ($N=999$, BH within tissue). Dual-null is the primary Support gate (both within-family BH $q_M < 0.05$ and $q_D < 0.05$). Dual $\wedge$ beats counts rows that are dual-null *and* have observed $\Delta\rho > 0$ (4/7 dual-null rows; 4/13 FM rows overall); shared-null significance is 2/13 (brain/PBMC scGPT only).

Tissue	Readout	ρ_{FM}	$\Delta\rho$ vs base	Dual-null	Beats base	Dual $\wedge$ beats	q_{Δ} (shared)	Shared sig
Brain	Geneformer embed	+0.00441	+0.00165	yes	yes	yes	0.379	no
Brain	scFoundation encoder	+0.00056	-0.00220	no	no	no	1.000	no
Brain	UCE encoder	+0.00456	+0.00180	yes	yes	yes	0.379	no
Brain	scGPT encoder	+0.01238	+0.00962	yes	yes	yes	0.008	yes
Brain	Geneformer attention	-0.00353	-0.00629	yes	no	no	1.000	no
Brain	Random-init floor	+0.00188	-0.00088	no	no	no	1.000	no
Brain	Geneformer KO	-0.00110	-0.00386	no	no	no	1.000	no
Brain	Artifact-corrected KO	-0.00197	-0.00473	no	no	no	1.000	no
PBMC	Geneformer embed	+0.00425	-0.00272	yes	no	no	1.000	no
PBMC	Geneformer attention	-0.00018	-0.00715	no	no	no	1.000	no
PBMC	scFoundation encoder	+0.00112	-0.00586	no	no	no	1.000	no
PBMC	scGPT encoder	+0.01098	+0.00401	yes	yes	yes	0.040	yes
PBMC	UCE encoder	+0.00593	-0.00105	yes	no	no	1.000	no

strata (brain $\rho = 0.00891$ versus 0.00398; PBMC 0.02111 versus 0.00141; E5). These checks report observed stability and construct sensitivity; E3 did not recompute valid randomization nulls after TF subsampling, and E5 uses a detection-based operational label rather than a biological housekeeping annotation.

A separate sequence-opportunity audit (S1) applied a five-bin target-opportunity permutation and passed its null-calibration checks, but it is not the frozen v2 primary result: several row-level statistics and q-values differ under that stricter sensitivity design. We therefore report S1 as a sensitivity audit rather than claiming that the primary headline has been re-tested under S1. External ENCODE pan-biosample TF-occupancy edges on the frozen panel (coverage gate: 204 TFs, 114,467 positive edges) correlate with the pooled motif+ATAC proxy under a lightweight within-TF encode-shuffle null (brain Spearman $\rho = 0.081$, PBMC $\rho = 0.109$; plus-one $p = 0.005$ at $N = 199$; mean per-TF AP ≈ 0.51 –0.53 among TFs with ≥ 5 positives; details archived in the release capsule). That association bounds the proxy as binding-proximal structure; it does *not* convert FM dual-null Support into causal TF–target recovery. Figure 5 reports the concordance (attention-likeness) check: a co-expression-like readout should correlate positively with co-expression. This check does *not* adjudicate residual regulatory content on its own. Five of the six dual-null positive rows fail concordance: their FM–co-expression Spearman correlations are -0.18 to -0.02 (all negative). PBMC scGPT weakly passes (Spearman +0.01) yet still fails protocol-pass (e.g. non-degree same-sign). The open red ring in Fig. 5A is brain Geneformer attention: it passes concordance but is dual-null negative, so a concordance passer need not be dual-null supported (and conversely). Fig. 5C shows the Geneformer knockout (KO) raw and position-shift-corrected KO; Fig. 5D shows opposite signs for brain Geneformer embedding versus attention. We report the six positives as readout-specific geometry rather than as regulatory signal. Randomization and knockout diagnostics for the same rows are summarized in Fig. 6; per-TF embedding-norm / cosine anisotropy versus degree is *not* tested in this release and is not used as a mechanistic claim (SAP §8).

3.3 The alignments depend on conditioning on proxy-derived degree covariates

Removing the two degree covariates changes both effect direction and randomization behavior (Fig. 7). Dual-null Support falls from 7 under full to 0 under non-degree, with 6 sign flips

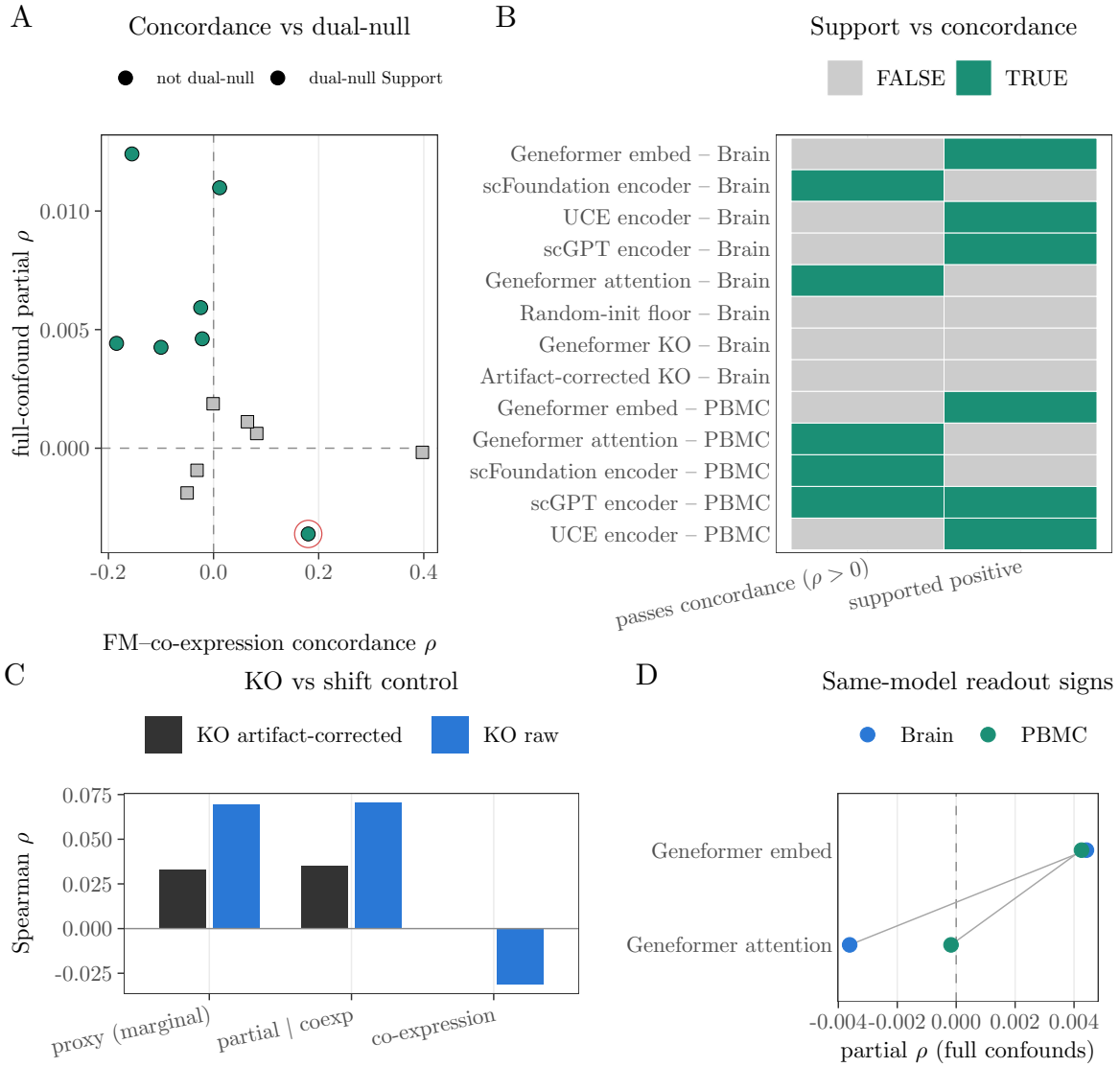


Figure 5: Readout concordance (attention-likeness) and internal consistency for foundation-model (FM) gene graphs. Concordance is the Spearman correlation of each FM graph with co-expression on the same edge set; it is an attention-likeness check, not a test of residual regulation. Dual-null Support requires both Benjamini–Hochberg q_M and q_D below 0.05. (A) Concordance versus full-confound partial ρ for all 13 pooled readouts; fill encodes dual-null Support, and the open red ring marks brain Geneformer attention (concordance passer, dual-null negative). (B) Positive dual-null Support and positive concordance as separate binary checks. (C) Geneformer knockout (KO) raw versus artifact-corrected (position-shift) KO, marginally and after partialling co-expression. (D) Same-model Geneformer embedding versus attention; brain signs are opposite.

(Fig. 7A–B); randomization p -values migrate upward (Fig. 7C), and the nested covariate ladder is a descriptive sequence of partial correlations rather than a model-selection exercise (Fig. 7D). No non-degree row is supported by both nulls, and three of the six full-spec positives (brain Geneformer embedding, PBMC Geneformer embedding, PBMC scGPT) are negative before degree conditioning. Because the degree covariates are computed from the proxy graph itself, the full specification conditions on functions of the outcome; the positives therefore appear in exactly the specification that carries this hazard. We report the non-degree specification as co-primary and treat the divergence as a substantive finding, not a footnote. Under non-degree, the audit records **0 dual**, **0 M-only**, and **9 D-only** rejections (row-shuffle $q_D < 0.05$ with gene-label $q_M \geq 0.05$); the gene-label null never rejects alone. Dual-Support is therefore never

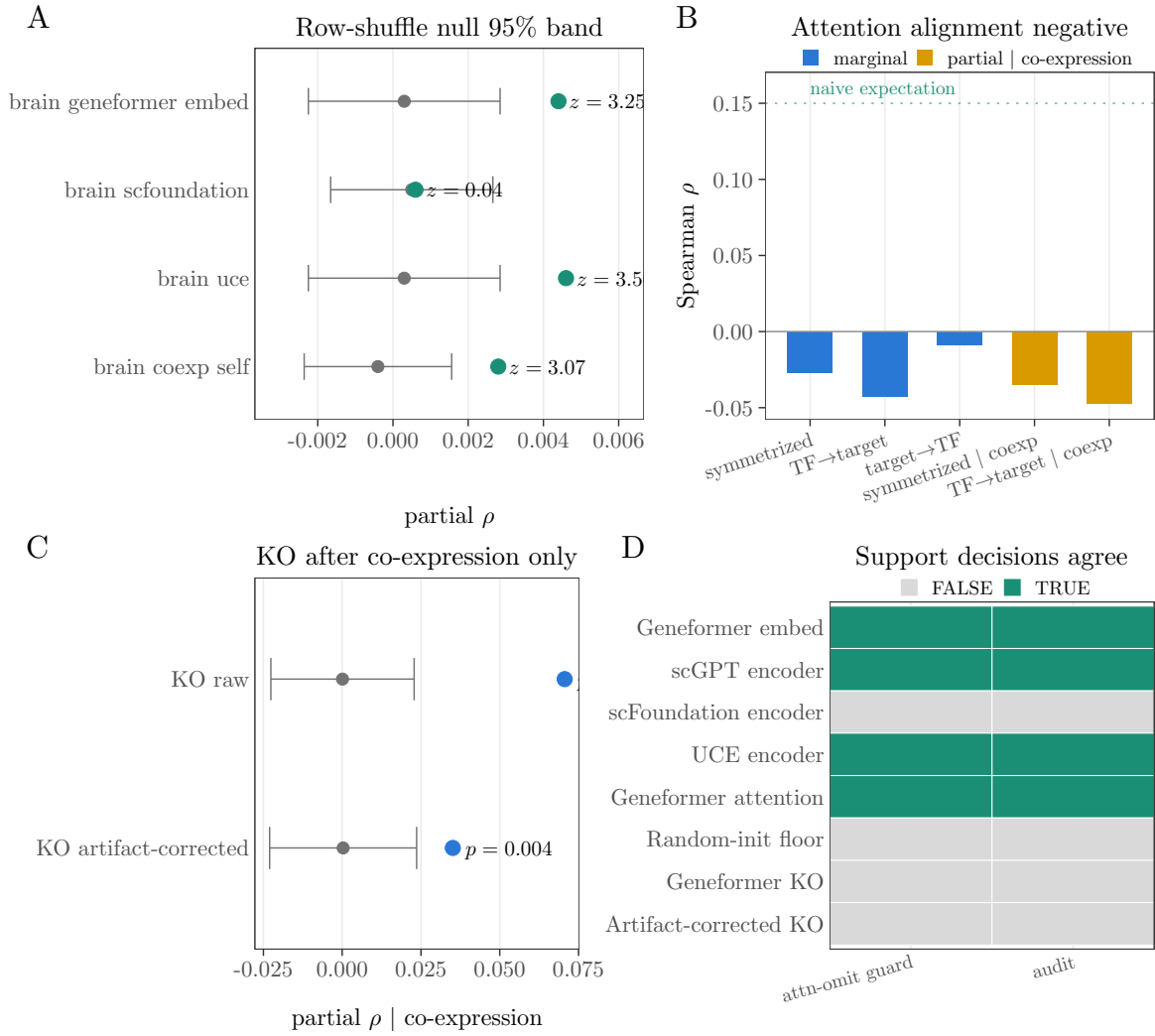


Figure 6: Randomization and readout diagnostics ($N = 999$ permutations). (A) Observed partial ρ (aqua) against the within-TF-row degree-preserving row-shuffle null mean and 95% null band (grey) for representative rows; z is (observed – null mean)/null SD. (B) Brain Geneformer-attention alignment, marginally and after adjusting for co-expression; the dotted “naive expectation” line is $\rho = 0.15$, not a fitted target. (C) Geneformer knockout (KO) raw and artifact-corrected (position-shift) KO both align with the proxy after partialling co-expression alone ($p < 0.001$ and $p = 0.004$); the full confound specification removes that alignment. (D) Dual-null Support decisions agree between the authoritative audit and a parallel attention-omission guard on the same eight brain rows (aqua = supported).

granted under non-degree. Those D-only rejections must not be read as dual-null Support: the row-shuffle still preserves TF out-degree while the non-degree design does not condition on degree, so the two nulls do not address the same estimand (SAP §3; release-capsule non-degree null-pattern summary). Table 7 enumerates that non-degree pattern row by row (0 dual; D-only majority).

3.4 Per-cell-type patterns are descriptive

The proxy itself is near cell-type-invariant on this panel (pairwise consensus $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast; we report them only as descriptive robustness summaries. The full-confound cell-type analysis contains 15 brain (3 readouts $\times$ 5 cell types) and 14 PBMC (2 readouts $\times$ 7 cell types) FM rows per specification (Fig. 8). Brain full-confound estimates range from -0.00396 to 0.00445

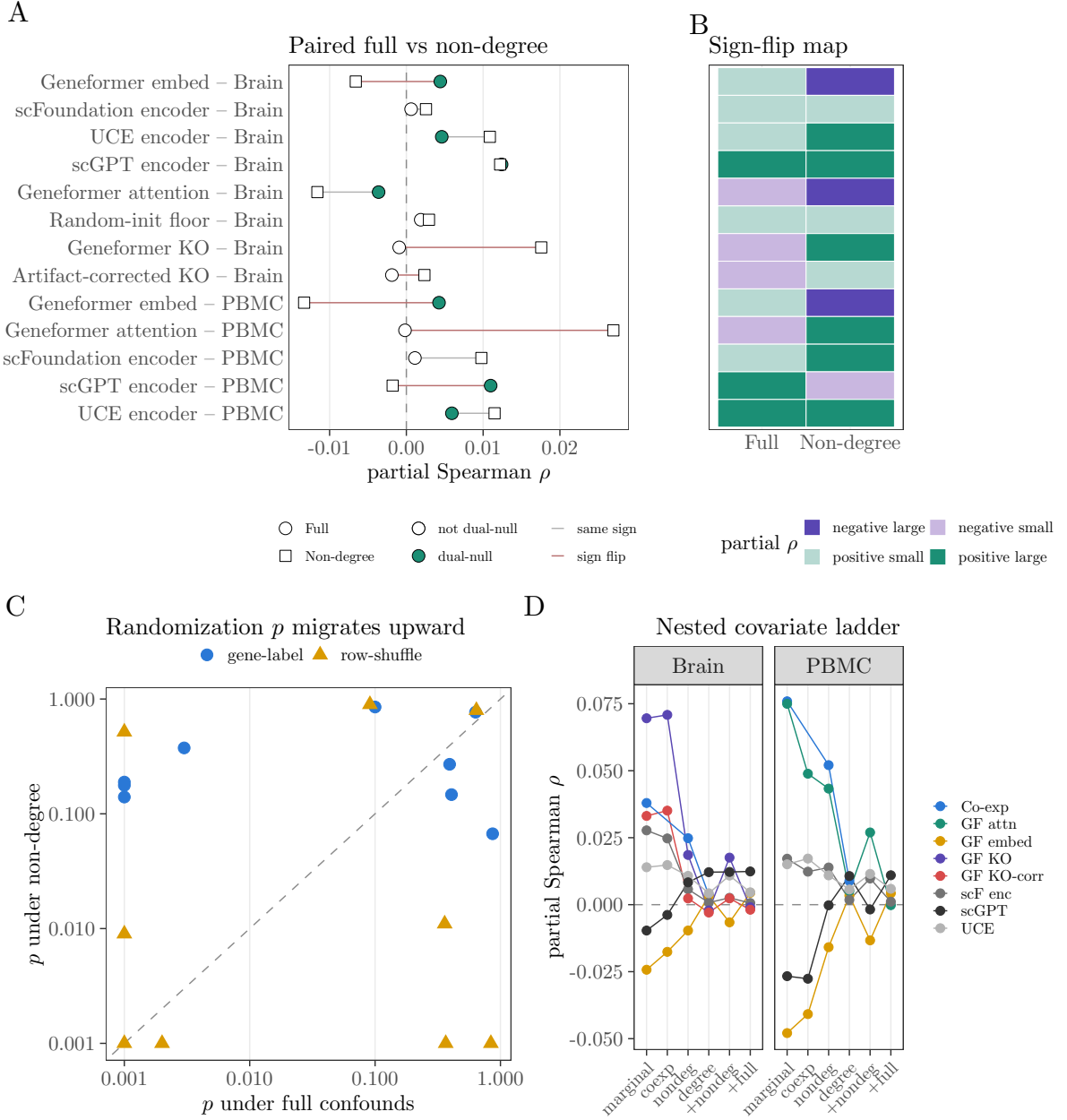


Figure 7: Dependence of the 13 foundation-model (FM) alignments on the confound specification. Full includes co-expression plus six structural covariates (including two proxy-derived degree terms); non-degree drops those two degree terms. Dual-null Support is $q_M < 0.05$ and $q_D < 0.05$. (A) Paired full (circles) versus non-degree (squares) partial ρ ; aqua fill = dual-null Support (7 under full, 0 under non-degree); red segments = sign flips (6/13). (B) Binned partial ρ (not a binary sign map): negative large / negative small / positive small / positive large at ± 0.005 . (C) Full versus non-degree randomization p for the gene-label and row-shuffle nulls. (D) Nested covariate ladder from marginal correlation through co-expression, non-degree, and degree rungs to the full specification; descriptive, not model selection.

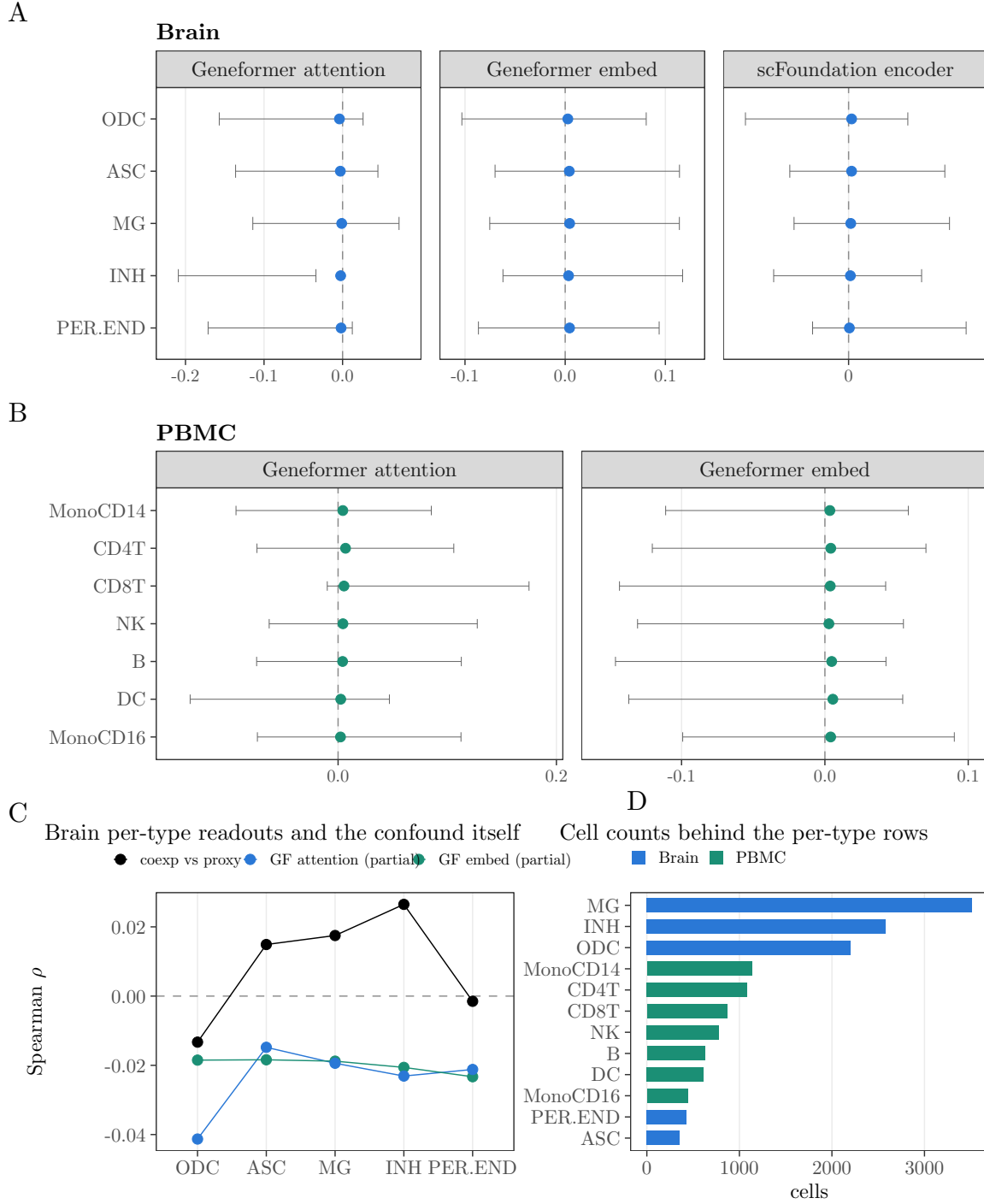


Figure 8: Per-cell-type descriptive robustness summaries of observed partial Spearman ρ under the full confound specification (brain: 15 exploratory FM $\times$ type rows; PBMC: 14). (A) Brain rows by readout (x = partial ρ ; y = cell type). (B) PBMC rows by readout. Whiskers in (A–B), where drawn, are bootstrap 95% intervals shown only as descriptive spread; they are not randomization p -values or population-uncertainty intervals. (C) Brain per-type co-expression, Geneformer-embedding, and Geneformer-attention summaries on the same frozen panel (legacy per-type readout overlay, not a second primary family). (D) Cell counts underlying the per-type rows. No multiplicity claims.

with median 0.00258; PBMC estimates range from 0.00222 to 0.00682 with median 0.00413. Table 6 summarizes full and non-degree ranges. These rows have no randomization p -values or BH correction. They neither support cell-type discoveries nor provide population uncertainty.

Table 6: Per-cell-type descriptive ranges of observed partial ρ under full and non-degree confound specifications (brain: 15 exploratory FM $\times$ type rows; PBMC: 14). Min/max/median across those rows only; no randomization p -values, BH correction, or population-uncertainty interpretation.

Tissue	Confound spec	n_{rows}	ρ_{min}	ρ_{max}	ρ_{median}
Brain	full	15	-0.00396	0.00445	0.00258
Brain	non-degree	15	-0.00986	0.01625	-0.00053
PBMC	full	14	0.00222	0.00682	0.00413
PBMC	non-degree	14	-0.01303	0.05371	0.01502

Table 7: Non-degree dual-null pattern for all 13 FM rows. The co-primary non-degree specification is the full design minus the two proxy-derived degree covariates. M is the gene-label permutation null and D is the within-TF-row degree-preserving target shuffle ($N = 999$ each); each q is BH-adjusted within the non-degree families. Dual-null Support ($q_M < 0.05$ and $q_D < 0.05$) is 0/13. Pattern “D-only” is $q_D < 0.05$ with $q_M \geq 0.05$; “neither” is both $q \geq 0.05$. Dual non-deg is reported in the protocol-pass matrix but is not required for protocol-pass.

Tissue	Model	ρ	q_M	q_D	Pattern
Brain	Geneformer embed	-0.00659	0.593	0.689	neither
Brain	scFoundation encoder	+0.00255	0.852	0.897	neither
Brain	UCE encoder	+0.01087	0.354	0.003	D-only
Brain	scGPT encoder	+0.01222	0.354	0.018	D-only
Brain	Geneformer attention	-0.01159	0.593	0.003	D-only
Brain	Random-init floor	+0.00294	0.354	0.045	D-only
Brain	Geneformer KO	+0.01757	0.354	0.003	D-only
Brain	Artifact-corrected KO	+0.00235	0.852	0.897	neither
PBMC	Geneformer embed	-0.01332	0.328	0.002	D-only
PBMC	Geneformer attention	+0.02696	0.328	0.002	D-only
PBMC	scFoundation encoder	+0.00977	0.338	0.014	D-only
PBMC	scGPT encoder	-0.00178	0.856	0.998	neither
PBMC	UCE encoder	+0.01149	0.328	0.002	D-only

3.5 The implementation responds monotonically to aligned injection, and the observed effects sit at the bottom of that ladder

The axis-aligned diagnostic rises monotonically from approximately zero at $\alpha = 0$ to mean partial correlations of 0.905 in brain and 0.913 in PBMC at $\alpha = 1$ (Fig. 9). The original ladder probes $\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}$; the later $\alpha \in \{0.002, 0.005, 0.01\}$ points are subdivided diagnostics used only to locate the observed magnitudes on that curve. They are not direct tests at each row’s inferred equivalent and do not constitute a power analysis, MDE, confidence interval, or exclusion bound.

3.6 A supervised TF-disjoint probe gives mixed family-level results

A linear probe trained to predict each TF’s proxy targets from graph features on 311 held-in TFs and tested on 135 disjoint TFs (Fig. 10) gives mixed family-level results after confound adjustment. The probe estimand is supervised TF-disjoint recoverability on **paired PBMC multiome only**; the eight brain rows are not probed. Paired sign-flip testing against co-expression supports the UCE contrast ($q = 0.0325$) and the random-init floor contrast ($q = 0.0300$) when the Benjamini–Hochberg family includes the floor, while Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not supported. As a

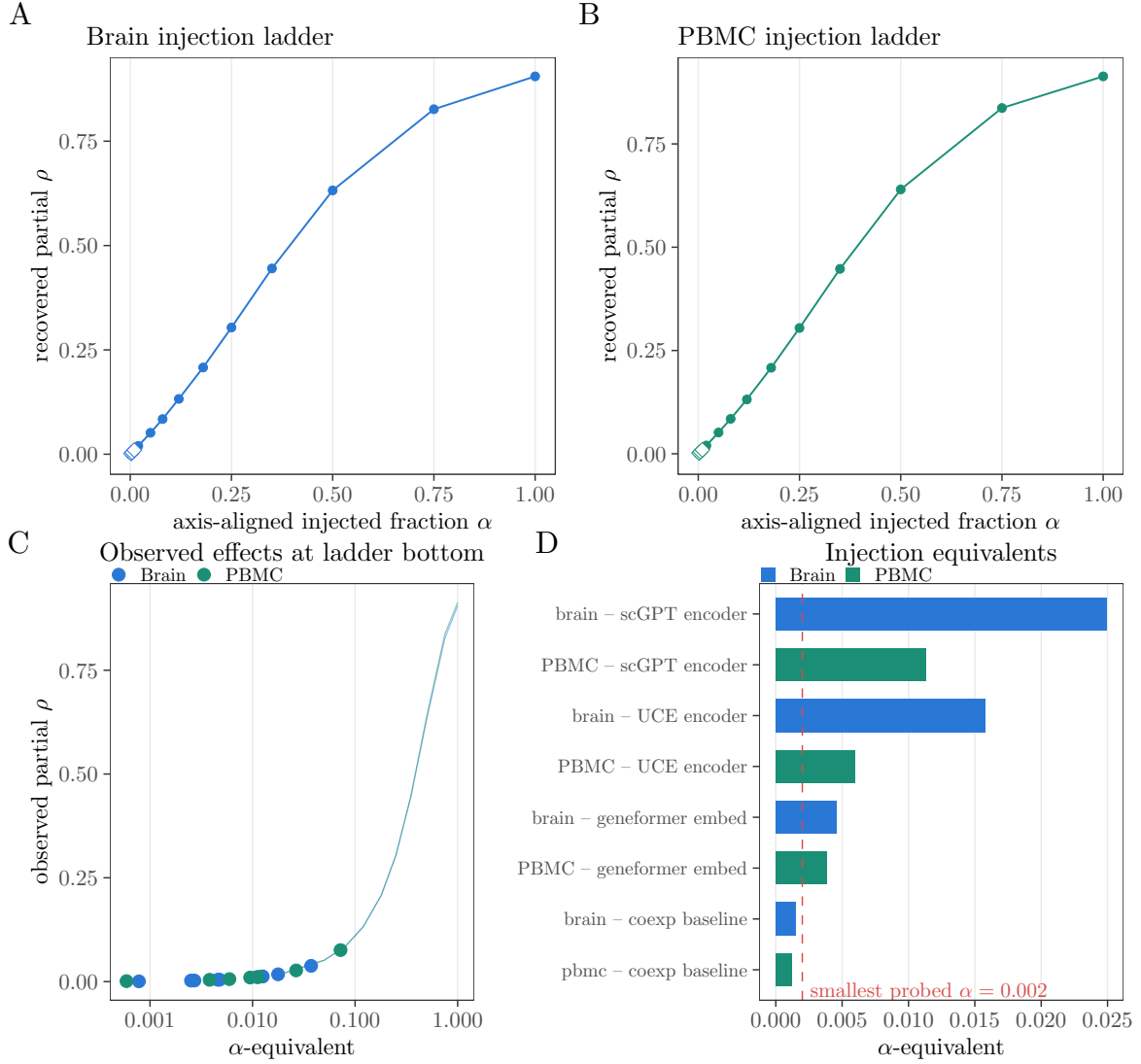


Figure 9: Axis-aligned pipeline-sensitivity diagnostic. Here α is the injected fraction of the proxy residual mixed with independent Gaussian noise. (A–B) Brain and PBMC ladders: lines are the mean recovered partial ρ across 30 independent noise replicates, ribbons show the replicate range, and diamonds show the subdivided low- α points. (C) Observed effects located on those injection curves. (D) α -equivalents of the six dual-null supported *positive* rows versus the smallest probed nonzero $\alpha = 0.002$; the negative brain Geneformer-attention row has no positive α -equivalent. This is not a confidence interval, power curve, minimum detectable effect, or exclusion analysis.

sensitivity check that treats the floor as a control rather than a peer FM, BH recomputed on the $n = 4$ FM contrasts *excluding random_floor* moves UCE to $q = 0.052$ (no longer < 0.05) with the other three FM contrasts remaining non-significant (release-capsule no-floor probe-contrast sensitivity). The sign-flip statistic compares how much each family is over-corrected, not how much it recovers: adjusted test correlations are essentially zero for UCE (+0.0003) and the floor (+0.0023) but negative for co-expression (−0.0120), so “supported” contrasts mean the baseline is over-corrected further below zero, not that UCE carries more residual regulatory signal. The probe’s confound set differs from the audit’s—it contains neither co-expression nor the proxy-derived degree covariates—so it answers a different question (supervised recoverability, not raw-readout alignment) and neither confirms nor refutes the pooled rows. In particular, the floor result and the no-floor sensitivity prevent reading the probe as confirmatory evidence that FMs beat co-expression. Table 8 reports the corresponding numeric family-level contrasts (PBMC

only; brain omitted), including the no-floor BH column.

Table 8: TF-disjoint probe numeric contrasts on paired PBMC multiome only (eight brain rows omitted). A linear probe is trained on 311 TFs and tested on 135 disjoint TFs; ρ is the confound-adjusted mean test Spearman. $\Delta\rho$ vs base is versus co-expression. Paired sign-flip q_{flip} includes the random-init floor in the BH family ($n = 5$ FM contrasts + floor); the no-floor column recomputes BH on the $n = 4$ FM contrasts excluding the floor (UCE moves to $q = 0.052$). “Supported” contrasts mean the baseline is over-corrected further below zero, not that the FM recovers residual regulation.

Family	ρ	$\Delta\rho$ vs base	p_{flip}	q_{flip}	q (no floor)
Co-expression	-0.01204	–	< 0.001	0.0015	–
Geneformer embed	-0.01627	-0.00423	0.410	0.5125	0.5467
Geneformer attention	-0.01019	+0.00185	0.650	0.6500	0.6500
scGPT encoder	-0.01914	-0.00710	0.153	0.2550	0.3060
UCE encoder	+0.00032	+0.01236	0.013	0.0325	0.0520
Random-init floor	+0.00232	+0.01436	0.006	0.0300	–

3.7 The construct transfers across three independent ATAC datasets

The proxy was built independently in three datasets (brain GSE174367, PBMC 10x multiome, and the GSE206767 fibroblast reprogramming pool), and Fig. 11 quantifies how much of each dataset the construct actually uses and shares. The ± 2 kb linkage admits only a small fraction of each dataset’s peaks: 8,823 of 219,070 in brain (4.03%), 6,609 of 111,743 in PBMC (5.91%), and 11,614 of 275,448 in the fibroblast pool (4.22%)—the proximity-linkage limitation is therefore a three-tissue property, not a PBMC-specific one. On the shared non-self TF–gene edge set (534,754 edges), 38,082 edges carry nonzero weight in all three consensus proxies (24% of the 158,838-edge union), and the TF out-degree marginals are similar across tissues (median 129–171 supported targets per TF)—which is exactly the structure the additive decomposition of Fig. 3 attributes most cross-tissue agreement to. Cross-tissue rank agreement also tracks binary-support agreement (observed ρ 0.462–0.525 versus support φ 0.446–0.507), so the construct transfers chiefly at the level of which edges exist, not of their continuous weights. The fibroblast arm carries no foundation-model readouts and enters none of the audit’s supported rows; it is a construct-level check, not evidence about model capability.

Independent ATAC overlays on DESCARTES spleen, bone-marrow mononuclear cells (BMMC; GSE194122), and a regulatory T-cell (Treg) ATAC pilot (GSE211155) transfer onto the three locked reference graphs at Mantel ρ 0.391–0.890, with BMMC–PBMC the strongest pair ($\rho = 0.890$) and spleen–PBMC the weakest; spleen agreement is largely additive (0.824–0.846) while BMMC–PBMC is not (additive fraction 0.460; Fig. A1). Projecting CollecTRI signed transcription-factor–gene interactions (26) onto the same frozen panel keeps 4,592/43,536 raw edges (0.105) and 327/446 TFs, about $19\times$ sparser than motif-binary support (85,809 edges), so the prior cannot replace the accessibility/motif graph (Fig. A2). Residual BMMC–PBMC rank match remains high after additive removal ($\rho = 0.950$), yet BMMC misses a predeclared 0.90 panel-replicate gate (1,076/1,200 genes, 341/446 TFs) and the Treg overlay is a filename-meta pilot (Fig. A3); these overlays do not add foundation-model (FM) audit rows.

Coverage, cell counts, and edge accounting for the same frozen panel are collected in Fig. 13.

4 Discussion

The audit finds six small positive fixed-panel alignments and one negative one, all supported by both randomization nulls under the full confound specification. Three features of that

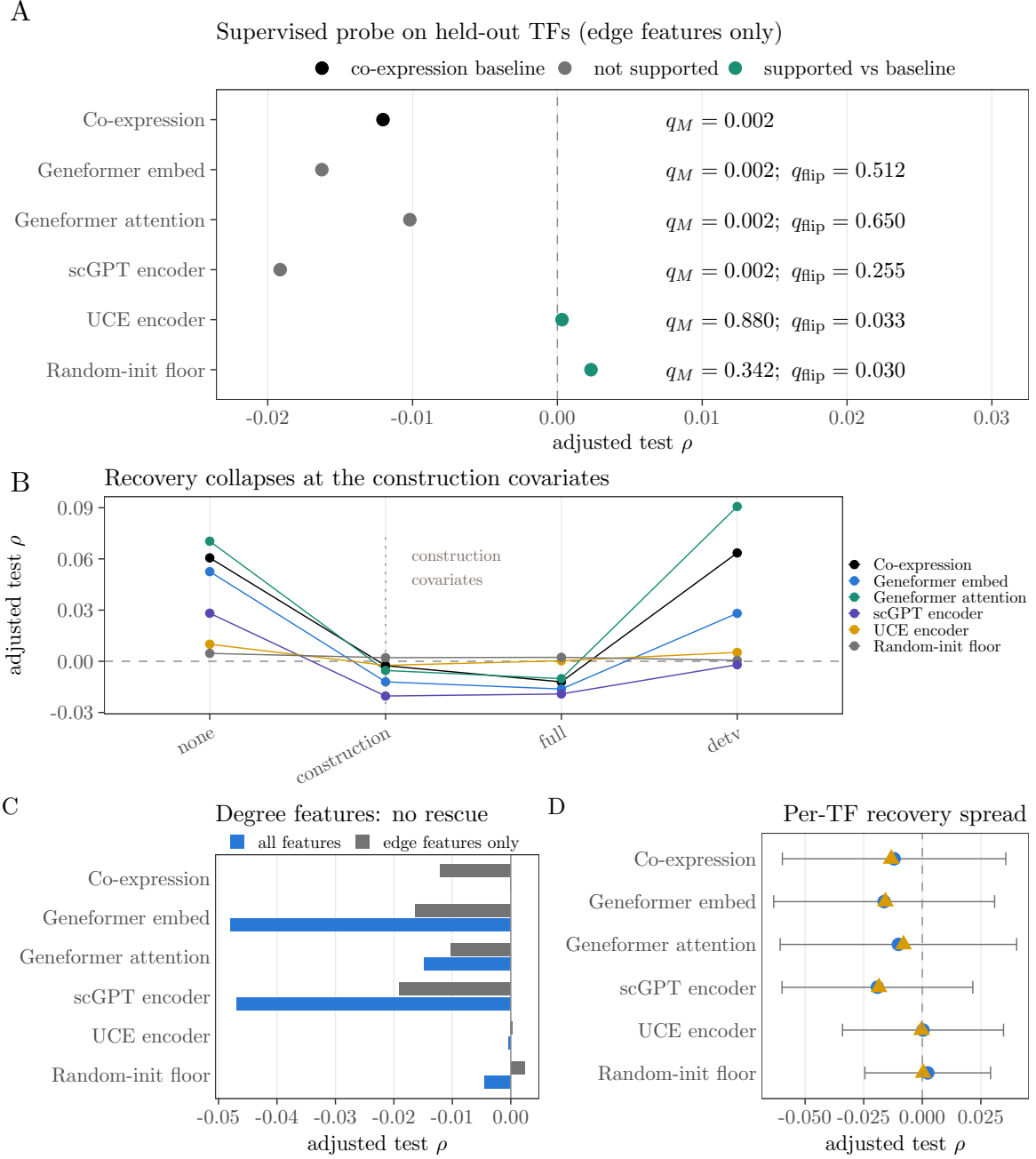


Figure 10: TF-disjoint supervised probe on paired PBMC multiome only (brain rows not probed; train $n = 311$ TFs, test $n = 135$). (A) Confound-adjusted test Spearman per family; labels are gene-label q_M and, for each FM family, paired sign-flip q_{flip} against co-expression. Aqua rows are supported against the baseline at $q_{\text{flip}} < 0.05$ (UCE 0.0325; random-init floor 0.0300) when the floor is in the BH family. (B) Adjusted Spearman across confound subsets (none, construction, full, detection-only); recovery collapses at construction covariates. (C) Edge-only primary arm versus the all-feature sensitivity arm. (D) Across-TF mean (circle), median (triangle), and ± 1 SD (whiskers) of adjusted Spearman. Supported probe contrasts are not residual-regulation claims.

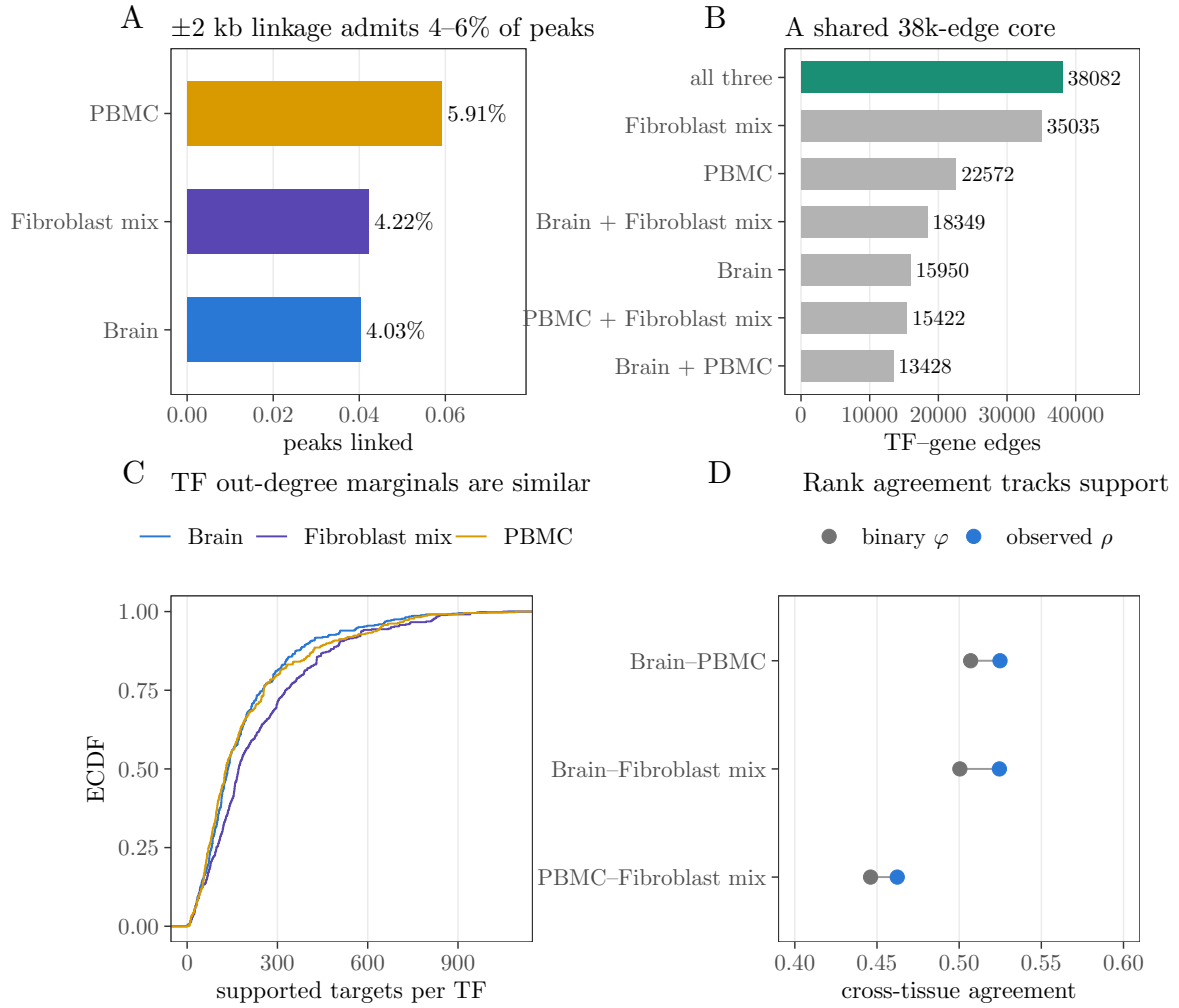


Figure 11: Third-tissue transfer of the proxy construct on brain (GSE174367), PBMC 10x multiome, and the fibroblast mix (GSE206767; construct-only, no FM rows). (A) Fraction of each dataset’s peaks admitted by the promoter-plus-gene-body ± 2 kb window (linked over total peaks). (B) Overlap of supported (nonzero) non-self TF-gene edges; aqua bar = 38,082-edge core present in all three. (C) Empirical CDF of TF out-degree (supported targets per TF; $n_{\text{TF}} = 446$) per tissue. (D) Observed cross-tissue Spearman ρ versus binary-support φ (phi of nonzero-edge presence/absence) for the three pairs. Descriptive rank/support agreement only; not a causal TF-target test and not an FM audit.

result restrain its interpretation. First, the co-expression baseline meets both uncorrected null thresholds on the same edge set, and a shared-null Mantel test of $\Delta\rho = \rho_{\text{FM}} - \rho_{\text{baseline}} > 0$ supports multiplicity-controlled superiority for only 2/13 rows (brain and PBMC scGPT; Table 5). Four dual-null positives beat their tissue baseline on the observed contrast; the rest—including PBMC Geneformer embedding—do not. Dual-null Support therefore does not imply separation from the co-expression confound under the frozen $\Delta\rho$ gate. Second, the two supported readouts of the same model in the same tissue (Geneformer embedding +0.0044, Geneformer attention -0.0036, both brain) have opposite signs, which is not the signature of a model-level regulatory capability. Third, three of the six positives are negative before conditioning on the proxy’s own degree covariates, so they appear only in the specification that conditions on functions of the outcome. A supervised TF-disjoint probe, answering the distinct question of supervised recoverability on paired PBMC data (brain rows are not probed), finds mixed contrasts: when the BH family includes the random-init floor, UCE and the floor are supported ($q = 0.0325$, $q = 0.0300$); excluding the floor as a peer FM moves UCE to $q = 0.052$. Adjusted test correlations are near zero for UCE/floor and more negative for co-expression, so

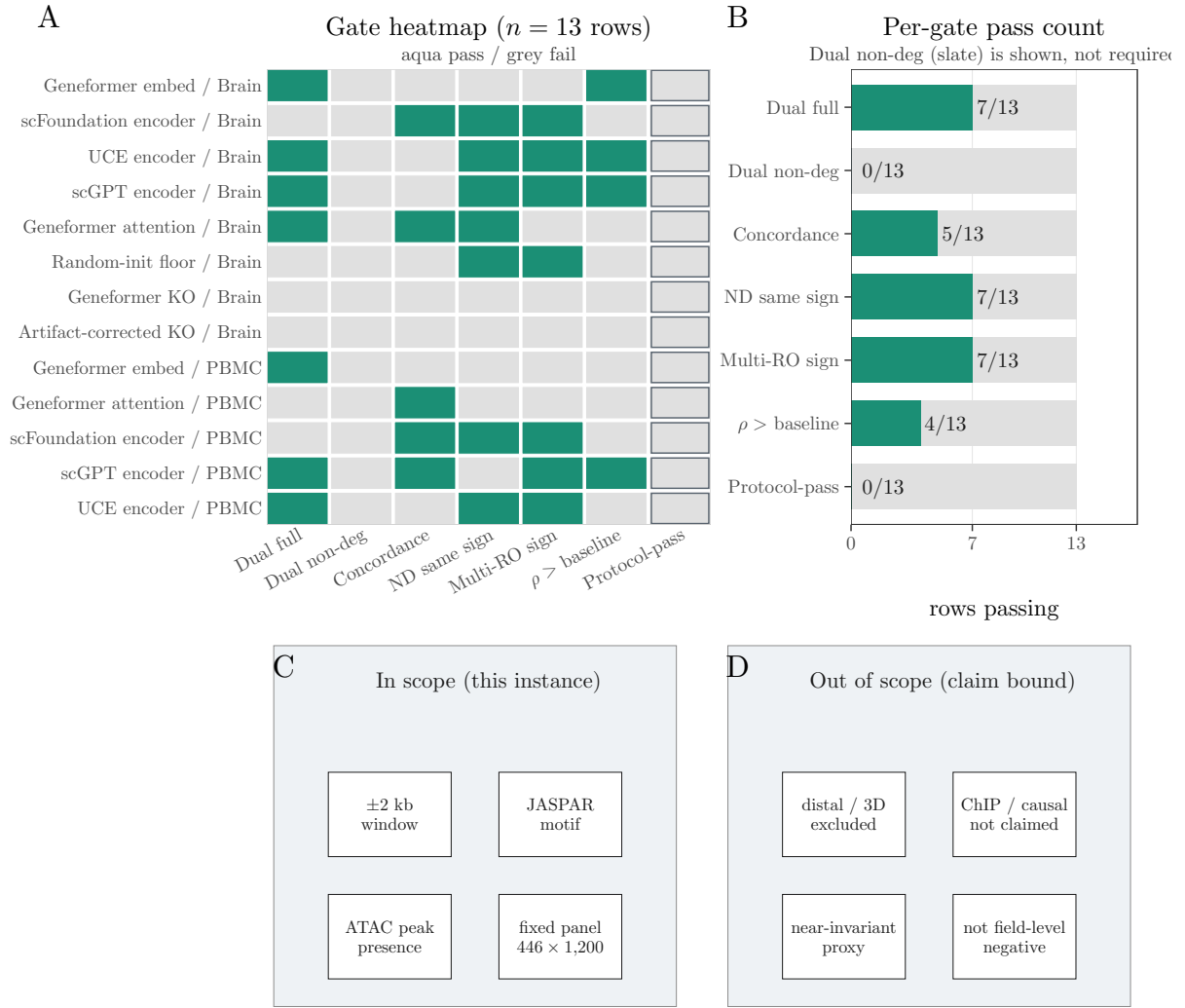


Figure 12: Protocol-pass gates and this audit instance’s scope on the thirteen pooled foundation-model (FM)/readout rows. (A) Binary gate heatmap (aqua pass, grey fail). Dual full / Dual non-deg: both within-family BH $q_M, q_D < 0.05$ under the named confound specification. Concordance: FM-co-expression Spearman > 0 . ND same sign: full and non-degree partial ρ share sign. Multi-RO sign: all full-spec readouts of the same model family in that tissue agree in sign (single-readout families pass). $\rho > \text{baseline}$: full-spec ρ exceeds the tissue co-expression baseline on the same edge set. Protocol-pass (outlined column) is the conjunction of Dual full, Concordance, ND same sign, Multi-RO sign, and $\rho > \text{baseline}$; Dual non-deg is shown but not required. No row passes (0/13). (B) Column sums of panel A: how many of the thirteen rows pass each gate (slate bar = Dual non-deg, reported only). (C) What this instance scores: JASPAR motif matches in ± 2 kb-linked ATAC peaks on the frozen $446 \times 1,200$ TF $\times$ gene panel. (D) What it does not score: distal/3D structure and ChIP-derived causal TF–target maps; the proxy is cell-type near-invariant on this panel. Failure on this instance cannot be read as a field-level negative.

“support” marks relative over-correction of the baseline, not recovered regulatory signal.

The most defensible reading is not that four models encode weak regulation. It is that small, readout-specific, specification-dependent alignments exist on this panel; that protocol-pass is 0/13; and that under the frozen estimand (Sec. 2.1) dual-null Support does not establish residual regulatory content after named confounds—especially once shared-null $\Delta\rho$ superiority is limited to 2/13 rows. A further observation points the same way: our own protocol freezes a concordance (attention-likeness) check in which a co-expression-like readout should correlate with co-expression (attention $\approx$ co-expression is well established). Five of the six supported pos-

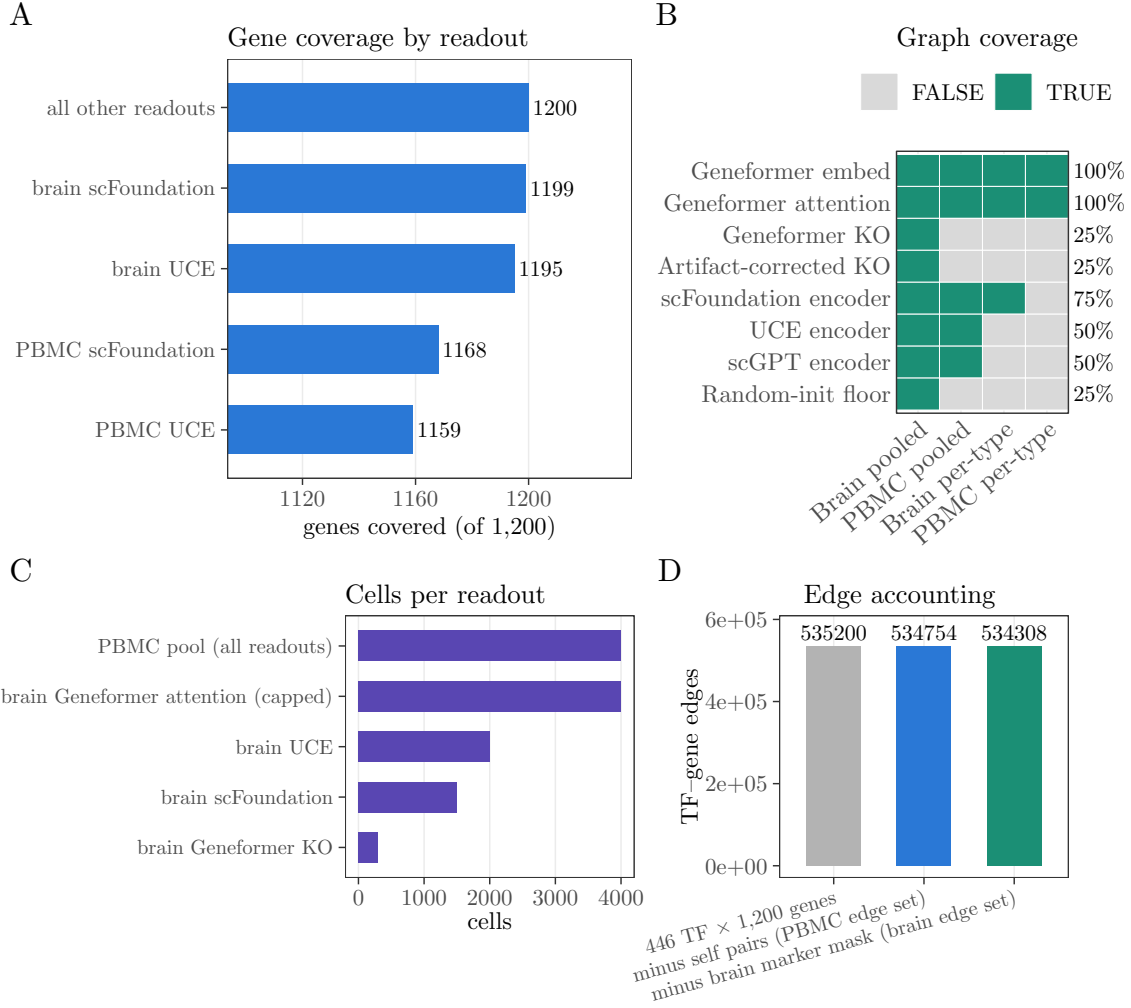


Figure 13: Coverage and fixed-panel quality control on the frozen $446 \times 1,200$ TF $\times$ gene panel. (A) Fraction of each encoder’s manifest genes present in the audit gene list (count out of 1,200). (B) Validated graph availability by tissue and analysis level (aqua = present, grey = absent); coverage is disclosed rather than assumed balanced. (C) Cell counts used by representative pooled inference runs. (D) Edge accounting from the Cartesian panel through self-pair exclusion (PBMC edge set) and the brain marker mask (brain edge set). This figure is a coverage disclosure, not a proof of balanced design.

itives fail that check (FM-co-expression Spearman -0.18 to -0.02), while concordance passers include PBMC scGPT ($+0.01$; still fails protocol-pass), scFoundation ($+0.08$), and PBMC Geneformer attention ($+0.40$)—the latter two are dual-null negative—so concordance does not adjudicate residual regulatory content, and dual-null Support without protocol-pass is not recovery. Embedding-cosine anisotropy versus degree remains an *untested* mechanistic hypothesis in this release (no frozen anisotropy diagnostic); we do not use it to explain dual-null positives. The pattern is compatible with reports that scFM-derived gene relationships often resemble co-expression (11, 12), and with zero-shot and biology-driven benchmarks that find scFMs unreliable or only weakly structured relative to simpler baselines (6, 7, 10). Failure to support a global regulatory-capability claim on this panel equally cannot support a field-level negative conclusion: the probe is linear and operates on pairwise graph summaries, the proxy is a narrow accessibility/motif construct (Fig. 12B), and other panels, tissues, or GRN-informed models could differ. Models pretrained with explicit GRN inductive biases (16, 17, 27) are outside the present multi-readout audit and remain a natural target for the same fixed-panel protocol.

Three statistical caveats apply throughout. First, one proxy randomization per replicate is

shared across the rows of a family, so the Monte Carlo p -values within a family are positively correlated; Benjamini–Hochberg remains valid under positive regression dependence (28), but the family-level error rate is then a joint property of the shared randomization—rejections occur across rows that share replicates, so the guarantee is weaker than the independent-per-row reading, not stronger. Second, in the non-degree specification the row-shuffle null preserves degree while the statistic does not condition on it, so the two nulls ask different questions there: several non-degree rows are unusual only under the row shuffle (brain Geneformer attention at $q_D = 0.003$ and the random-init floor at $q_D = 0.045$), which we read as evidence that degree adjustment—not regulatory content—drives that null’s rejections. Under non-degree, dual-null Support is 0/13 with 9 D-only rejections (Table 7); we do not treat D-only as Support. Third, replication depth differs by model: scGPT is supported in both tissues but not by the probe; UCE is supported in both tissues but its probe contrast depends on the family boundary; Geneformer is supported only through one of two readouts and its other readout is opposite-signed. We therefore report readout-level, not model-level, support.

Limitations and scope. The reference is a regulatory-potential proxy built from motif and accessibility evidence, and its coverage is narrow by construction. Panel membership further requires ≥ 1 brain ATAC peak in the linkage window (Sec. 2.2); a no-peak-filter manifest under the same $n = 1200$ cap overlaps the frozen panel in 1167/1200 genes (Jaccard 0.946) and enlarges the pre-cap universe from 12,382 to 12,690 genes (release-capsule no-ATAC-filter panel-sensitivity analysis). A full dual-null re-audit on that alternate universe was not re-run (would require new proxy and FM graphs). The ± 2 kb promoter-plus-gene-body rule retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%); distal enhancer elements, which carry most cell-type-specific TF occupancy (29), are systematically excluded, and no co-accessibility or 3D-genome linker (for example Cicero (30) or activity-by-contact models (31)) is used. At the MOODS threshold used here ($p < 10^{-5}$, 500 bp accessible windows, both strands, 554 motifs), the expected random hit count is 5.5 per peak while the observed counts are 20.5 (brain), 23.2 (PBMC), and 21.7 (fibroblast mix), so roughly a quarter of motif hits are expected false positives (32); no information-content threshold, conservation filter, or PWM-length correction (33) is applied. The panel is also composition-biased: 30 TFs in brain (31 in PBMC) share an identical binary target profile with at least one other TF (five FOX-family members are one such group), 128 of 446 TFs in brain (140 in PBMC) have a partner at binary cosine > 0.8 , and in the reference lineage lists we checked, 7 of 21 neural TFs are present while all 37 housekeeping TFs are, because the panel cap ranks candidates by RNA detection rate. The proxy is near cell-type-invariant on this panel (pairwise $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast. Brain analyses combine unpaired snATAC-seq with cross-study RNA, whereas the PBMC data are paired multiome, so the co-expression control is intrinsically stronger in PBMC than in brain. The full specification conditions on proxy-derived degree covariates; the non-degree co-primary specification avoids that hazard and supports no row under both nulls. The panel is fixed rather than sampled, so p_{MC} describes unusualness under the stated randomizations, not uncertainty over a TF or tissue population; no effect size carries an interval estimate, and with $N \approx 5.3 \times 10^5$ edges, effects of $|\rho| \approx 0.002$ are detectable, so statistical support alone cannot distinguish a small effect from no biologically meaningful effect. With $N = 999$ permutations, 0.001 is the resolution floor of every Monte Carlo p -value, not an exact value; rows with zero null exceedances are reported as $p < 0.001$. The supervised TF-disjoint probe is run on paired PBMC multiome data only; it does not directly test the eight brain rows, whose RNA and ATAC measurements come from unpaired sources. Per-cell-type and cross-tissue analyses are descriptive. The injection diagnostic is axis-aligned and cannot establish power against arbitrary biological alternatives. Fine-tuning, few-shot adaptation, decoder outputs, non-linear probes on raw per-gene embeddings, other gene panels, and direct perturbational validation are not tested. To make the path

to broader validation explicit, we have inventoried and registered two additional single-cell RNA collections for follow-up replication: a development collection (27 datasets, $\approx 2.35 \times 10^6$ cells, mixed human and mouse) and a cancer collection (28 datasets, $\approx 1.20 \times 10^6$ cells, predominantly human), with per-file accession, cell count, and modality recorded in the project registry. These collections are RNA-only, so they cannot drive the accessibility/motif proxy used here; they are reserved for a planned co-expression-side replication that reuses the same fixed panel and confound specification once an orthogonal binding reference (for example ENCODE TF ChIP-seq) is in place.

The same accessibility/motif construct on independent spleen, BMMC, and Treg ATAC still agrees with the locked references in the Mantel ρ 0.39–0.89 range, so cross-ATAC transfer is not confined to the three main-text tissues (Fig. A1). A literature prior (CollecTRI) is too sparse on this panel to serve as a substitute reference (Fig. A2), and disclosed coverage and metadata gates keep BMMC below a 0.90 replicate threshold and Treg as a filename-meta pilot (Fig. A3). Those panels bound how the proxy moves with new ATAC; they are not scored against FM graphs in this article.

5 Conclusion

scReg-Eval is first a fixed-panel audit protocol and software capsule: hash-pinned panel and nulls, dual-null Support reporting, concordance checks, and baseline comparison on a shared edge set. Applied to thirteen pooled model/readout rows against an accessibility/motif regulatory-potential proxy, it finds six small positive and one negative dual-null alignment under the full confound specification, a co-expression baseline that reaches the same order of alignment, shared-null $\Delta\rho > 0$ significance for only 2/13 rows (brain and PBMC scGPT; Table 5), protocol-pass 0/13, sign-inconsistent readouts within the same model, and positives that depend on conditioning on outcome-derived covariates, of which five of six fail concordance (PBMC scGPT weakly passes yet fails protocol-pass). A supervised disjoint-TF probe gives mixed family-level contrasts, with the UCE and random-init floors distinguishing from co-expression but no model-specific regulatory recovery. These application-scene results neither support a global regulatory-capability claim for the audited graphs nor license a field-level negative conclusion beyond this protocol instance. The design complements living GRN benchmarks and standardized scFM task suites (5, 6) by building proxy edge weights without expression statistics while still disclosing panel and ATAC filters as expression-adjacent design choices.

6 Conflict of Interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 Author Contributions

CRedit author contributions: Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

8 Funding

This work received no dedicated or external funding. [Placeholder: replace with grant numbers if applicable.]

9 Data Availability Statement

Code, fixed-panel manifests, seed metadata, model/readout coverage tables, authoritative audit outputs, and figure sources are available at <https://github.com/PeterPonyu/scfm-reg-audit> and archived at <https://doi.org/10.5281/zenodo.21724336>. The repository and archive support three levels of reproducibility: (i) validation that every published number and hash can be checked without external downloads; (ii) statistical reruns of the fixed-panel audit against the released outputs; and (iii) a full-pipeline recipe with dataset accessions, checkpoint hashes, a locked environment, and ordered commands. Public inputs include GSE174367 brain data (34), the 10x Genomics 10k PBMC multiome (35), and the GSE206767 chemical-reprogramming ATAC pool. Appendix construct-lane overlays additionally use DESCARTES spleen ATAC, BMMC GSE194122, Treg GSE211155, and CollecTRI signed interactions (26). Large upstream datasets and model weights are not redistributed; the availability record documents their stable source identifiers and the exact manifests used. GSE206767 is a chemical-reprogramming experiment: it contains day-0 BJ fibroblasts and day-3 induced neuron-like cells, pooled here as one accessibility track because no per-barcode condition metadata is distributed with the matrix (36).

10 Ethics Statement

This study reuses public, de-identified datasets and performs no new human-subject recruitment or intervention. Ethics approval for the original data generation is as stated in the source publications and GEO records.

11 Declaration of generative AI use

A large language model was used as an auxiliary aid for limited engineering edits to analysis code. No AI system generated scientific data, randomization outputs, or any reported number. The author takes full responsibility for the manuscript and accompanying software. No AI tool is listed as an author.

References

- [1] Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, et al. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023. doi: 10.1038/s41586-023-06139-9.
- [2] Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- [3] Aditya Pratapa, Amogh P. Jalihal, Jeffrey N. Law, Aditya Bharadwaj, and T. M. Murali. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17(2):147–154, 2020. doi: 10.1038/s41592-019-0690-6.
- [4] Daniel Marbach, James C. Costello, Robert Küffner, Nicole M. Vega, Robert J. Prill, et al. Wisdom of crowds for robust gene network inference. *Nature Methods*, 9(8): 796–804, 2012. doi: 10.1038/nmeth.2016.
- [5] Jalil Nourisa, Antoine Passemiers, Jeremie Kalfon, Marco Stock, Berit Zeller-Plumhoff, Robrecht Cannoodt, et al. geneRNIB: a living benchmark for gene regulatory network

inference. *bioRxiv*, 2025. doi: 10.1101/2025.02.25.640181. Preprint; OpenProblems living GRN benchmark.

- [6] Ping Qiu, Qianqian Chen, Hua Qin, Shuangfang Fang, Yilin Zhang, et al. BioLLM: A standardized framework for integrating and benchmarking single-cell foundation models. *Patterns*, 6(8):101326, 2025. doi: 10.1016/j.patter.2025.101326.
- [7] Jialu Wu, Qing Ye, Yilin Wang, Renling Hu, Yiheng Zhu, et al. Biology-driven insights into the power of single-cell foundation models. *Genome Biology*, 26:334, 2025. doi: 10.1186/s13059-025-03781-6.
- [8] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8):1657–1661, 2025. doi: 10.1038/s41592-025-02772-6.
- [9] Rebecca Boiarsky, Nalini M. Singh, Alejandro Buendia, Gad Getz, and David Sontag. A deep dive into single-cell RNA sequencing foundation models. *bioRxiv*, 2023. doi: 10.1101/2023.10.19.563100. Preprint.
- [10] Kasia Z. Kedzierska, Lorin Crawford, Ava P. Amini, and Alex X. Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26:101, 2025. doi: 10.1186/s13059-025-03574-x. Article number 101; supersedes the 2023 bioRxiv preprint.
- [11] Ihor Kendiukhov. Systematic evaluation of single-cell foundation model interpretability reveals attention captures co-expression rather than unique regulatory signal. *arXiv*, 2026. Preprint.
- [12] Ihor Kendiukhov. Sparse autoencoders reveal organized biological knowledge but minimal causal regulatory logic in Geneformer and scGPT. *arXiv*, 2026. Preprint.
- [13] Minsheng Hao, Jing Gong, Xin Zeng, Chiming Liu, Yucheng Guo, Xingyi Cheng, et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21(8):1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- [14] Yanay Rosen, Yusuf Roohani, Ayush Agrawal, Leon Samotorčan, Tabula Sapiens Consortium, Stephen R. Quake, and Jure Leskovec. Universal cell embedding provides a foundation model for cell biology. *Nature*, 656(8126):183–191, 2026. doi: 10.1038/s41586-026-10689-z.
- [15] Fan Yang, Wenchuan Wang, Fang Wang, Yuan Fang, Duyu Tang, Junzhou Huang, et al. scBERT as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data. *Nature Machine Intelligence*, 4(10):852–866, 2022. doi: 10.1038/s42256-022-00534-z.
- [16] Xiaodong Yang, Guole Liu, Guihai Feng, Dechao Bu, Pengfei Wang, et al. GeneCompass: deciphering universal gene regulatory mechanisms with a knowledge-informed cross-species foundation model. *Cell Research*, 34(12):830–845, 2024. doi: 10.1038/s41422-024-01034-y.
- [17] Luni Hu, Hua Qin, Yilin Zhang, Yi Lu, Ping Qiu, et al. RegFormer: a single-cell foundation model powered by gene regulatory hierarchies. *Nature Communications*, 17, 2026. doi: 10.1038/s41467-026-72198-x.
- [18] Ieva Rauluseviciute, Rafael Riudavets-Puig, Romain Blanc-Mathieu, Jaime A. Castro-Mondragon, Katalin Ferenc, Vipin Kumar, et al. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52(D1):D174–D182, 2024. doi: 10.1093/nar/gkad1059.

- [19] Janne Korhonen, Petri Martinmäki, Cinzia Pizzi, Pasi Rastas, and Esko Ukkonen. MOODS: fast search for position weight matrix matches in DNA sequences. *Bioinformatics*, 25(23):3181–3182, 2009. doi: 10.1093/bioinformatics/btp554.
- [20] Sara Aibar, Carmen Bravo González-Blas, Thomas Moerman, Vân Anh Huynh-Thu, Hana Imrichova, Gert Hulselmans, et al. SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11):1083–1086, 2017. doi: 10.1038/nmeth.4463.
- [21] Hongzhi Wen, Wenzhuo Tang, Xinnan Dai, Jiayuan Ding, Wei Jin, Yuying Xie, and Jiliang Tang. CellPLM: Pre-training of cell language model beyond single cells. In *International Conference on Learning Representations (ICLR)*, 2024. Also available as bioRxiv 10.1101/2023.10.03.560734.
- [22] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1993. ISBN 978-0-412-04231-7.
- [23] Nathan Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2 Part 1):209–220, 1967. PMID: 6018555.
- [24] Belinda Phipson and Gordon K. Smyth. Permutation P-values should never be zero: Calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), 2010. doi: 10.2202/1544-6115.1585. Article 39.
- [25] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [26] Sophia Müller-Dott, Eirini Tsirvouli, Miguel Vazquez, Ricardo O. Ramirez Flores, Pau Badia-i Mompel, Robin Fallegger, Dénes Túrei, Astrid Lægreid, and Julio Saez-Rodriguez. Expanding the coverage of regulons from high-confidence prior knowledge for accurate estimation of transcription factor activities. *Nucleic Acids Research*, 51(20):10934–10949, 2023. doi: 10.1093/nar/gkad841.
- [27] Jérémie Kalfon, Jules Samaran, Gabriel Peyré, and Laura Cantini. scPRINT: pre-training on 50 million cells allows robust gene network predictions. *Nature Communications*, 16, 2025. doi: 10.1038/s41467-025-58699-1.
- [28] Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4):1165–1188, 2001. doi: 10.1214/aos/1013699998.
- [29] Robert E. Thurman, Eric Rynes, Richard Humbert, Jeff Vierstra, Matthew T. Maurano, et al. The accessible chromatin landscape of the human genome. *Nature*, 489(7414): 75–82, 2012. doi: 10.1038/nature11232.
- [30] Hannah A. Pliner, Jonathan S. Packer, Jose L. McFaline-Figueroa, Darren A. Cusanovich, Riza M. Daza, Delasa Aghamirzaie, Sanjay Srivatsan, Xiaojie Qiu, Dana Jackson, Anna Minkina, Andrew C. Adey, Frank J. Steemers, Jay Shendure, and Cole Trapnell. Cicero predicts cis-regulatory DNA interactions from single-cell chromatin accessibility data. *Molecular Cell*, 71(5):858–871.e8, 2018. doi: 10.1016/j.molcel.2018.06.044.
- [31] Charles P. Fulco, Joseph Nasser, Thouis R. Jones, Glen Munson, Drew T. Bergman, et al. Activity-by-contact model of enhancer–promoter regulation from thousands of CRISPR perturbations. *Nature Genetics*, 51(12):1664–1669, 2019. doi: 10.1038/s41588-019-0538-0.

- 723 [32] Wyeth W. Wasserman and Albin Sandelin. Applied bioinformatics for the identification of
724 regulatory elements. *Nature Reviews Genetics*, 5(4):276–287, 2004. doi: 10.1038/nrg1315.
- 725 [33] Gary D. Stormo. DNA binding sites: representation and discovery. *Bioinformatics*, 16(1):
726 16–23, 2000. doi: 10.1093/bioinformatics/16.1.16.
- 727 [34] Samuel Morabito, Emily Miyoshi, Neethu Michael, Saba Shahin, Alessandra Cadete
728 Martini, et al. Single-nucleus chromatin accessibility and transcriptomic characterization
729 of Alzheimer’s disease. *Nature Genetics*, 53(8):1143–1155, 2021. doi:
730 10.1038/s41588-021-00894-z.
- 731 [35] 10x Genomics. PBMC from a healthy donor, single cell multiome ATAC + gene
732 expression (10k cells). Cell Ranger ARC reference dataset, 2021. 10x Genomics support
733 reference multiome dataset.
- 734 [36] Gene Expression Omnibus. Chemical-based external stimulation reprograms BJ’s into
735 FiNs (scATAC-seq).
736 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE206767>, 2023. Accession
737 GSE206767; day-0 BJ fibroblasts and day-3 induced neuron-like cells; accessed 2026-07.

738 A Additional construct-lane ATAC overlays

739 The thirteen main-text figures report the fixed-panel scReg-Eval audit. The three figures below
740 add construct-lane ATAC from spleen, BMNC, and Treg on the same frozen panel; they are
741 additional data, not a restatement of the main-text proxy figure, not a substitute for Figures 1–
742 13, and they do not add foundation-model audit rows. Spearman ρ is the Mantel rank statistic
743 of Eq. (6) between two TF–gene matrices on $n_{\text{TF}} = 446$ shared transcription factors from the
744 frozen $446 \times 1,200$ gene panel. Additive fraction and residual ρ are Eq. (7).

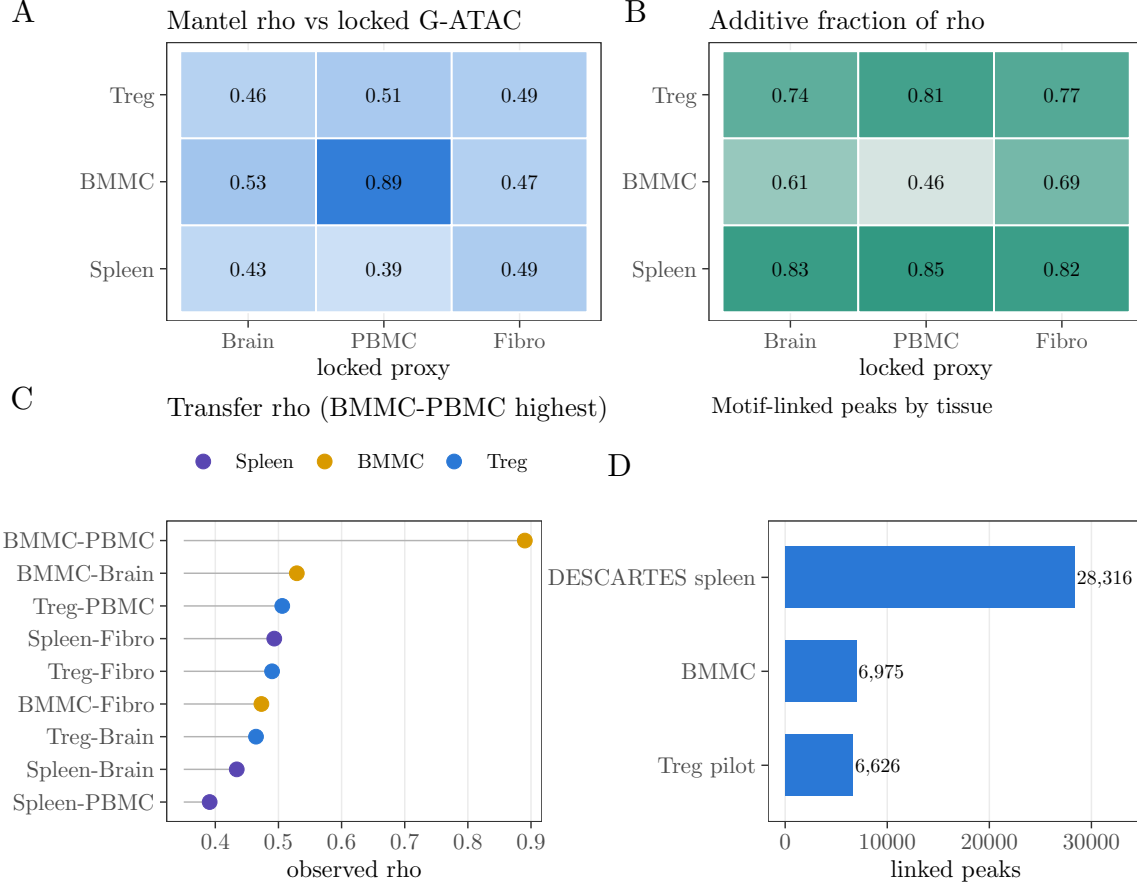


Figure A1: Additional construct-lane ATAC: Mantel transfer of the accessibility/motif transcription-factor–target proxy from spleen, BMMC, and Treg onto the three locked reference graphs. Each overlay graph is a $\text{TF} \times \text{gene}$ matrix built by scanning JASPAR motifs in accessible peaks and linking hits to genes by a promoter-plus-gene-body rule. Reference graphs use the same construct on adult brain (GSE174367), 10k PBMC multiome, and fibroblast/induced-neuron mix (GSE206767). Overlay tissues are DESCARTES spleen, bone-marrow mononuclear cells (BMMC; GSE194122), and a regulatory T-cell (Treg) ATAC pilot (GSE211155). All nine pairs are scored on $n_{\text{TF}} = 446$ shared TFs. **(A)** Observed Spearman ρ . Spleen versus brain/PBMC/fibroblast mix: 0.434, 0.391, 0.493. BMMC versus the same references: 0.529, 0.890, 0.473. Treg versus the same: 0.464, 0.506, 0.490. The BMMC–PBMC cell is the strongest transfer; spleen–PBMC is the weakest. **(B)** Additive fraction of that ρ (TF-row plus gene-column marginals). Spleen: 0.830, 0.846, 0.824. BMMC: 0.608, 0.460, 0.689. Treg: 0.742, 0.805, 0.766. Spleen agreement is largely additive construct machinery; BMMC–PBMC is not. **(C)** The same nine pairs ranked by observed ρ , coloured by overlay tissue. **(D)** Motif-linked peak counts used as construct input: spleen 28,316, BMMC 6,975, Treg 6,626. Peak depth does not monotone-explain transfer ρ . These quantities are descriptive rank agreement. They are not a causal TF–target recovery test and do not add foundation-model audit rows.

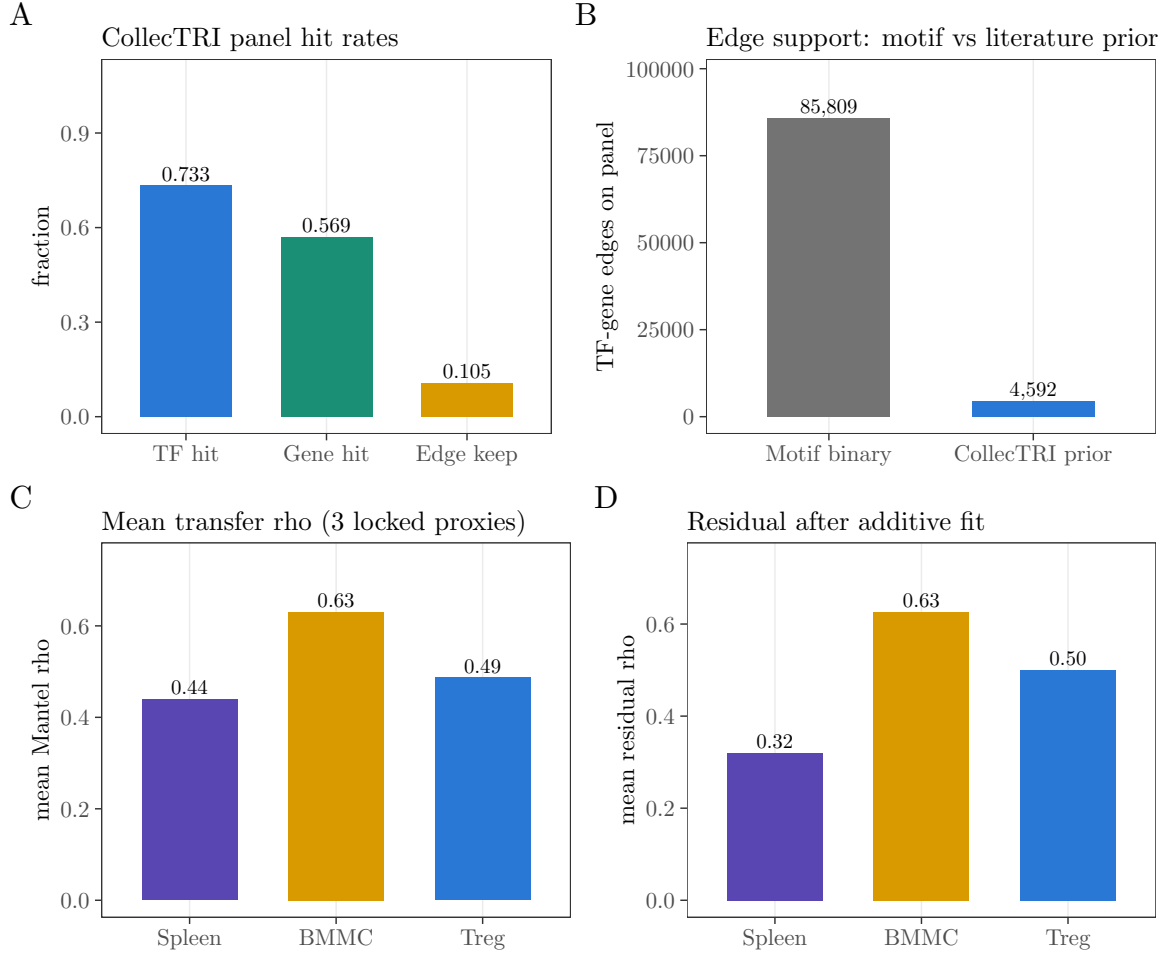


Figure A2: CollecTRI literature prior versus motif-binary support on the fixed panel, with summary transfer statistics. CollecTRI signed TF–gene interactions were projected onto the $446 \times 1,200$ panel (absolute weight; maximum over duplicate edges). Motif-binary support is the count of nonzero edges in a magnitude-ablated copy of the brain reference graph. **(A)** Hit rates: 327/446 TFs (0.733), 683/1,200 genes (0.569), and 4,592/43,536 raw CollecTRI edges kept (0.105). The prior is TF-rich relative to genes and sparse relative to its own table. **(B)** Edge counts on the panel: motif-binary 85,809 versus CollecTRI 4,592 (about 19 $\times$ sparser). A prior of this density cannot replace the accessibility/motif graph as a reference. **(C)** Mean Mantel ρ of each overlay against the three reference graphs: spleen 0.44, BMMC 0.63, Treg 0.49. **(D)** Mean residual ρ after additive removal: spleen 0.32, BMMC 0.63, Treg 0.50. BMMC’s residual mean matching its transfer mean is the low-additive-fraction fact in another view; spleen’s drop 0.44 $\rightarrow$ 0.32 is the complementary additive fact. CollecTRI is not treated as a causal regulon gold standard.

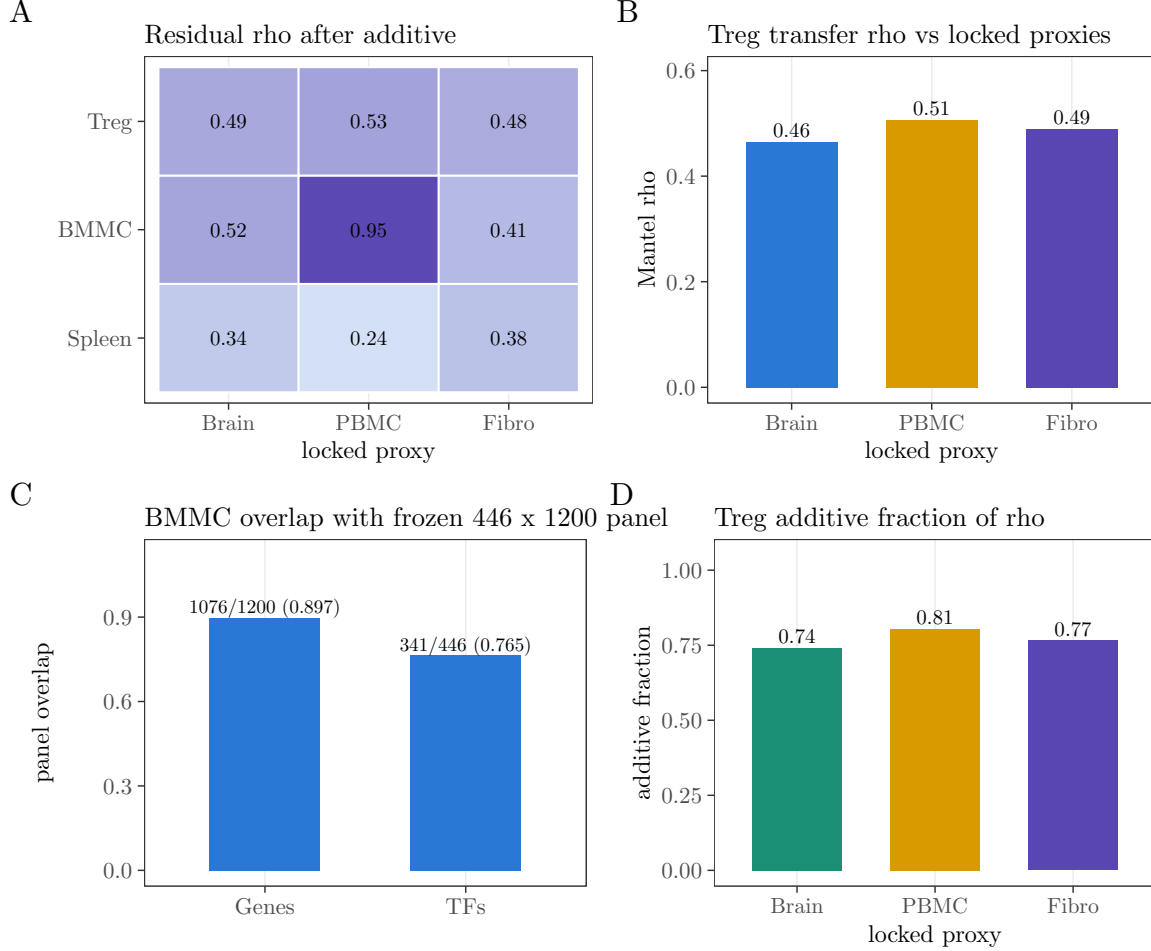


Figure A3: Residual rank match, Treg metadata limits, BMMC coverage against a 0.90 gate, and a blocked tumor ATAC input. **(A)** Residual Spearman ρ after removing additive TF-row and gene-column fits. Spleen residuals: 0.340, 0.236, 0.383 versus brain, PBMC, and fibroblast mix. BMMC residuals: 0.515, 0.950, 0.414. Treg residuals: 0.494, 0.525, 0.482. Residual ρ can exceed observed ρ when additive structure is net-subtractive (BMMC–PBMC: observed 0.890, residual 0.950). **(B)** Treg-pilot Mantel ρ versus the three reference graphs (0.464, 0.506, 0.490). The Treg population label is taken from filename-sorted metadata, not from recovered cell-type columns in the object `obs` field. A local set of 115 peak-matrix files had no cell-type-like `obs` columns, so this overlay is a filename-meta pilot. **(C)** BMMC overlap with the fixed $446 \times 1,200$ panel: 1,076/1,200 genes (0.897) and 341/446 TFs (0.765). Both miss a predeclared 0.90 replication gate, so BMMC is not treated as a complete panel replicate. **(D)** Treg additive fractions (0.742, 0.805, 0.766), showing a more additive profile than BMMC–PBMC. Human Tumor Atlas Network glioblastoma sample C3N-01334 was a candidate overlay and is blocked: only fragment files were available (no peak matrix). It is not a Mantel row and is not drawn as a bar. These panels disclose inclusion limits. They do not add foundation-model audit rows.